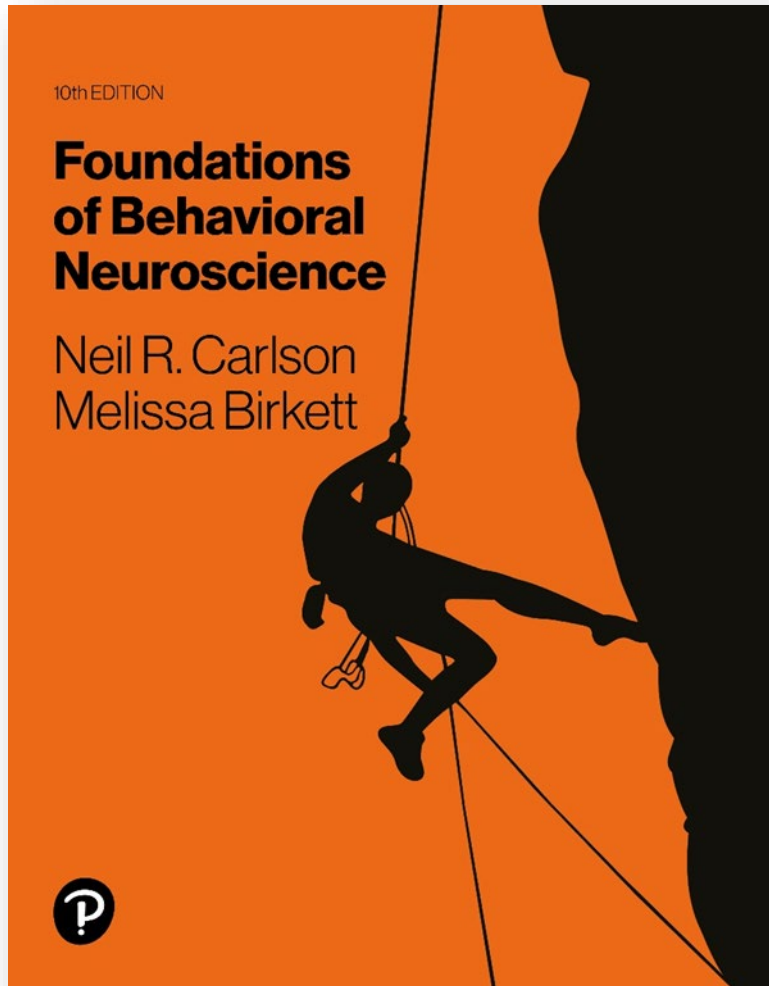


Foundations of Behavioral Neuroscience

Tenth Edition



Chapter 16

Schizophrenia, Affective Disorders, Anxiety Disorders, and OCD

Schizophrenia

- **Description**
- **Genetic Factors**
- **Environmental Factors**
- **The Mesolimbic Dopamine Pathway: Positive Symptoms**
- **The Mesocortical Dopamine Pathway: Negative and Cognitive Symptoms**

Schizophrenia

Schizophrenia is probably the most misused psychological term in existence

Description (1 of 2)

Learning Objective 16.1 Describe the positive, negative, and cognitive symptoms of schizophrenia.

Positive Symptoms

- Presence of unusual behaviors, in excess of typical functioning
 - **Thought disorders**
 - **Delusions**
 - **Hallucinations**

Description (2 of 2)

Negative Symptoms

- Absence or decrease in some typical behaviors
 - Social withdrawal, lack of affect, and reduced motivation
- Cognitive Symptoms
 - Cognitive deficits, such as difficulty sustaining attention, low psychomotor speed, deficits in learning and memory, poor abstract thinking, poor problem solving

What other disorder do these symptoms sound like?

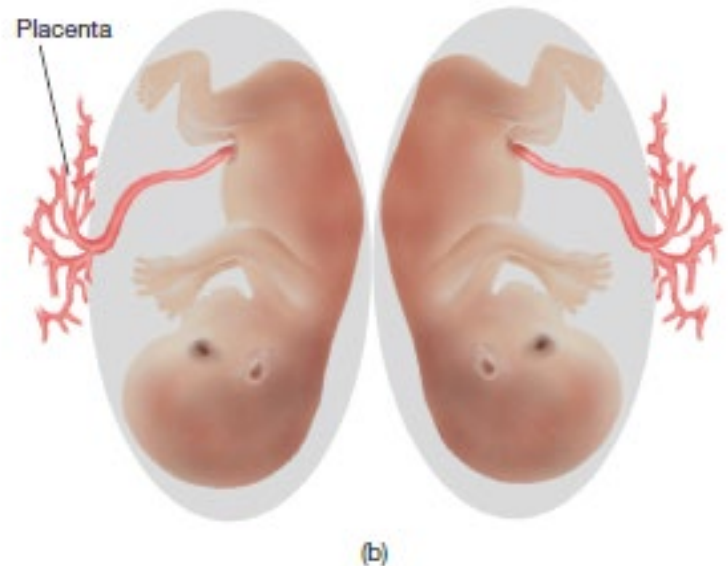
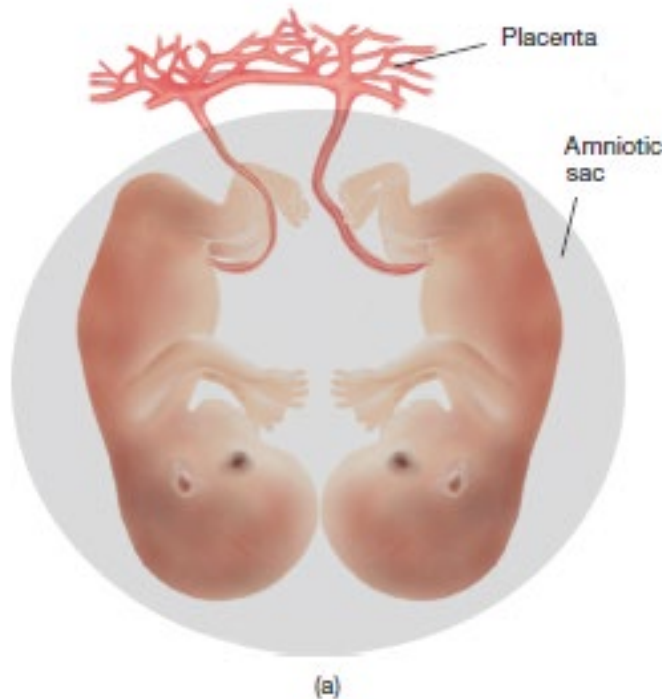
Description (2 of 2)

- Symptoms must persist for a length of time
- Rule out other explanations, such as alcohol abuse, bipolar disorder, depression
- Typical onset is during late adolescence and early adulthood

Genetic Factors (2 of 3)

Twin and Adoption Studies

- Studies indicate heritability
- Higher rate in monochorionic monozygotic twins



Genetic Factors (1 of 3)

Learning Objective 16.2 Describe genetic factors associated with the development of schizophrenia.

Factors

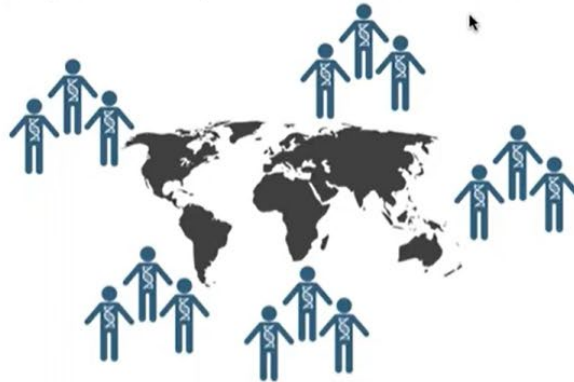
- Heritability plays a role, but no “schizophrenia gene”
- Evidence of susceptibility to develop schizophrenia

Mutations

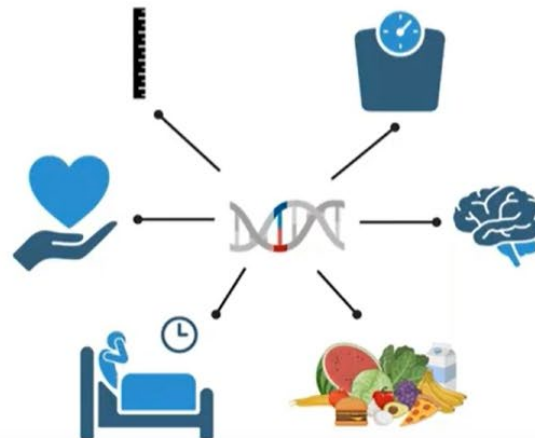
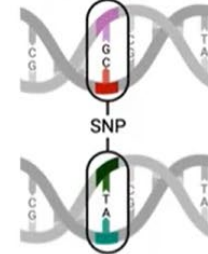
- Mutation of gene known as DISC1 increases likelihood of schizophrenia and other mental disorders

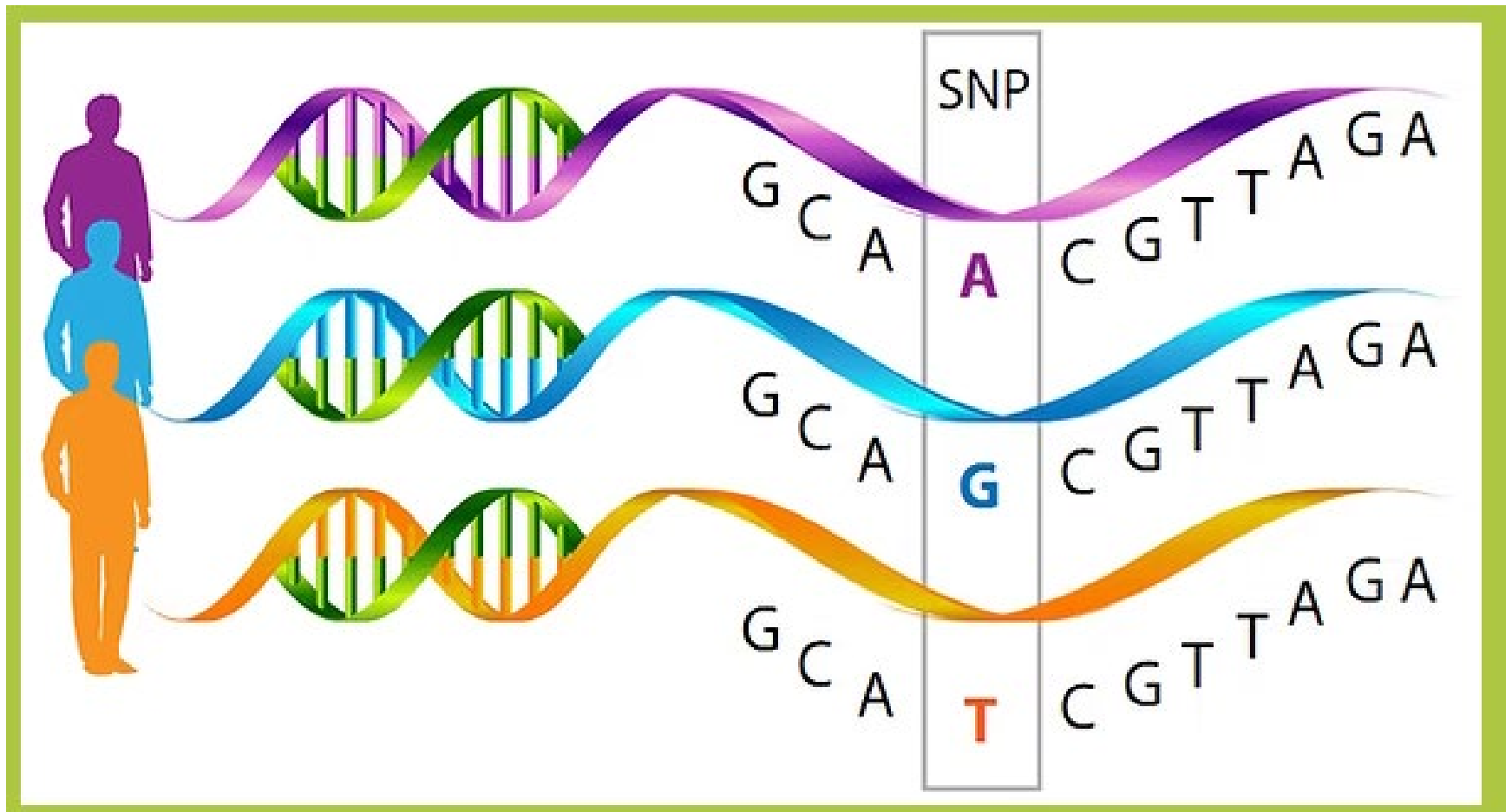
Genetics Primer

Our genetic makeup is 99.9% identical to one another



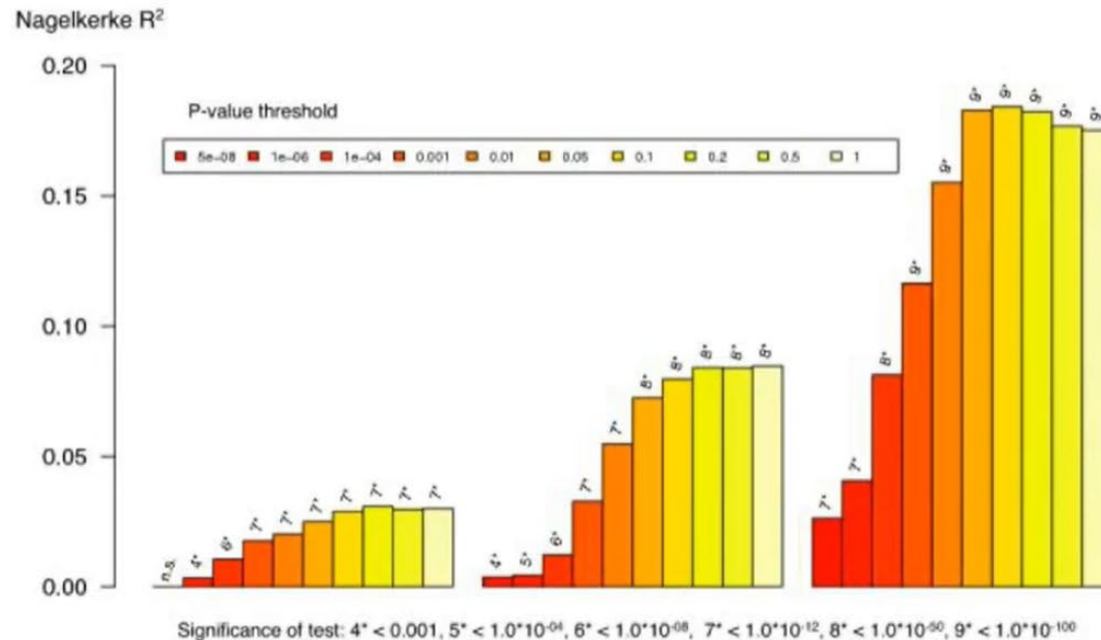
<0.01% of differences (things that make us unique) are what we are studying





Genetic risk for schizophrenia

- Not one, but **many** genes used to calculate risk



Ripke et al., 2014

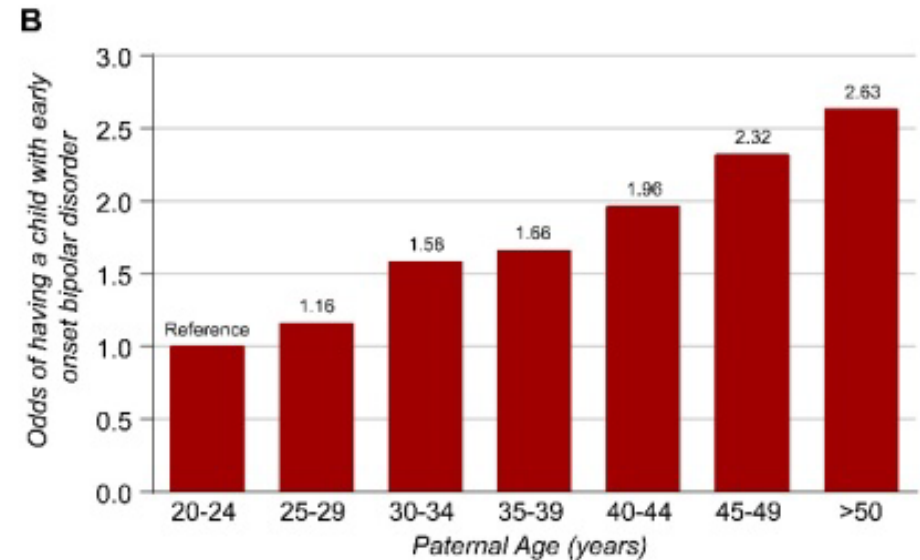
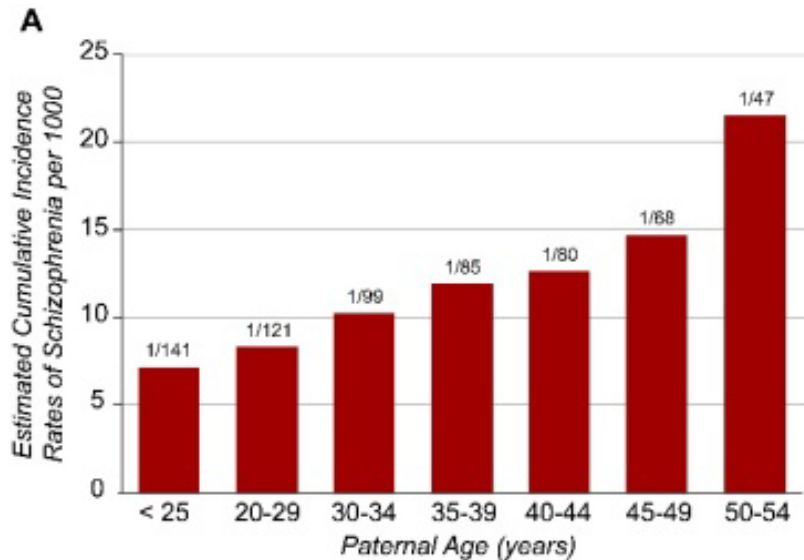
Genetic Factors (3 of 3)

Paternal Age

- Children of older fathers are more likely to develop schizophrenia
 - Increased likelihood of mutations in chromosomes of cells that produce sperms

Epigenetics

- Rare mutations or epigenetic factors may predispose people to schizophrenia
- Methylation leads to suppression of gene



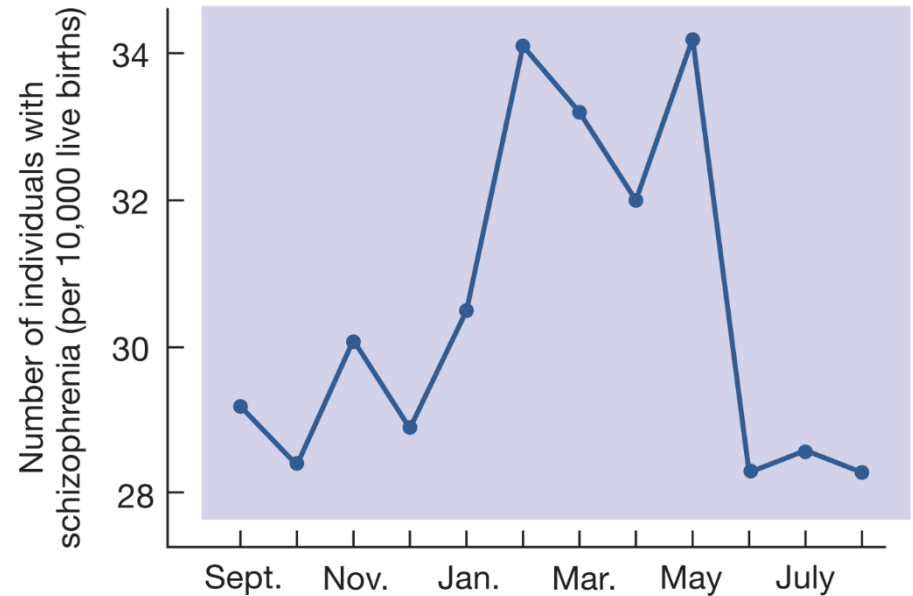
- Feingold KR, Anawalt B, Blackman MR, et al., editors. South Dartmouth (MA): MDText.com, Inc.; 2000-.

Environmental Factors

Learning Objective 16.3 Describe environmental factors associated with the development of schizophrenia.

Associated Factors

- Season of Birth
- Viral Epidemics
- Vitamin D Deficiency
- Population Density
- Prenatal Stress
- Substance Abuse



The Mesolimbic Dopamine Pathway:

Positive Symptoms (1 of 2)

Learning Objective 16.4 Provide evidence of the role of the mesolimbic dopamine pathway in the positive symptoms of schizophrenia.

Effects of Dopamine Agonists and Antagonists

- Dopamine hypothesis: positive symptoms of schizophrenia are caused by hyperactivity of dopaminergic synapses in the mesolimbic pathway
- Drugs acting as agonists produce and reinforce positive symptoms

D2 receptors and PET scan

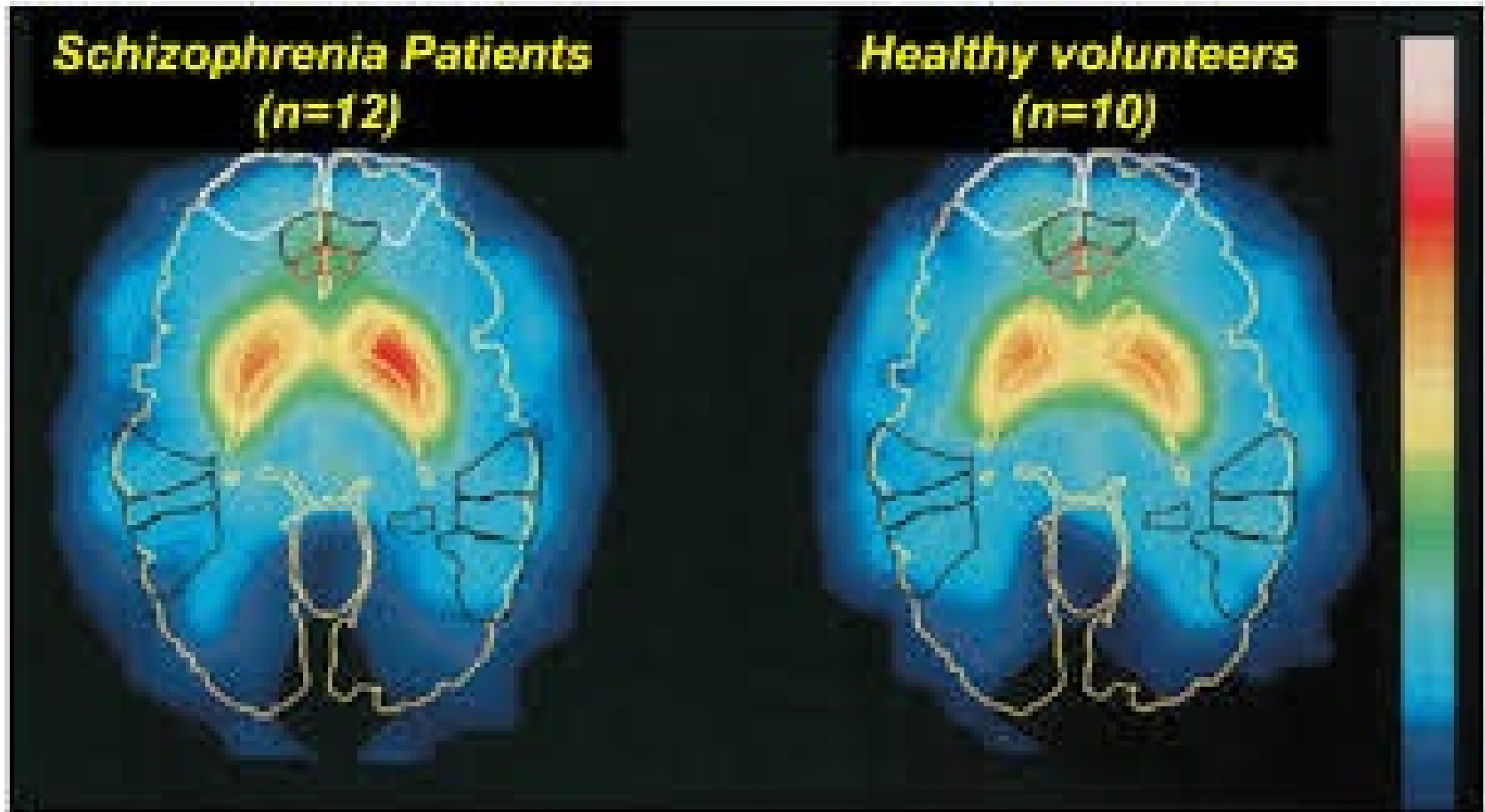
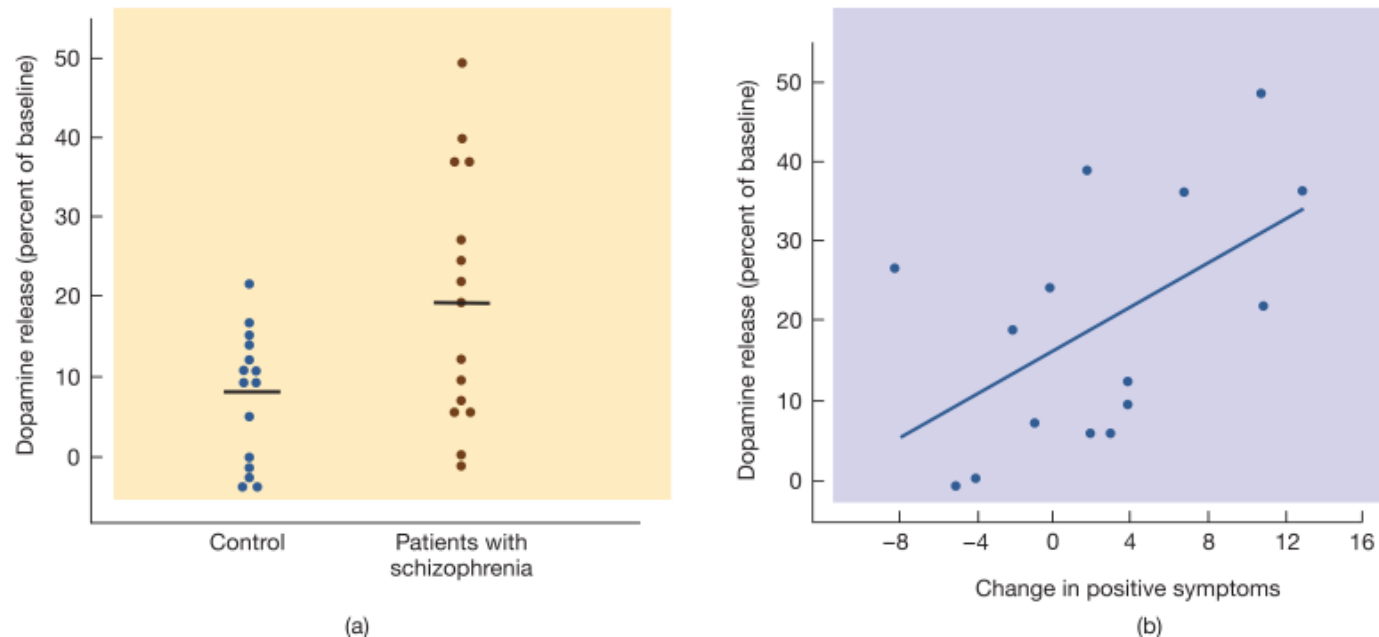


Figure 16.3

Increased Dopamine Activity Is Associated with Positive Symptoms of Schizophrenia



(a) Relative amount of dopamine released in response to amphetamine. (b) Relationship between dopamine release and changes in positive symptoms among patients with schizophrenia. Source: (Based on Laruelle, M., Abi-Dargham, A., Van Dyck, C. H., Gil, R., D'Souza, C. D., Erdos, J., . . . & Baldwin, R. M. (1996). Single photon emission computerized tomography imaging of amphetamine-induced dopamine release in drug-free schizophrenic subjects. *Proceedings of the National Academy of Sciences*, 93(17), 9235–9240.)

The Mesolimbic Dopamine Pathway:

Positive Symptoms (2 of 2)

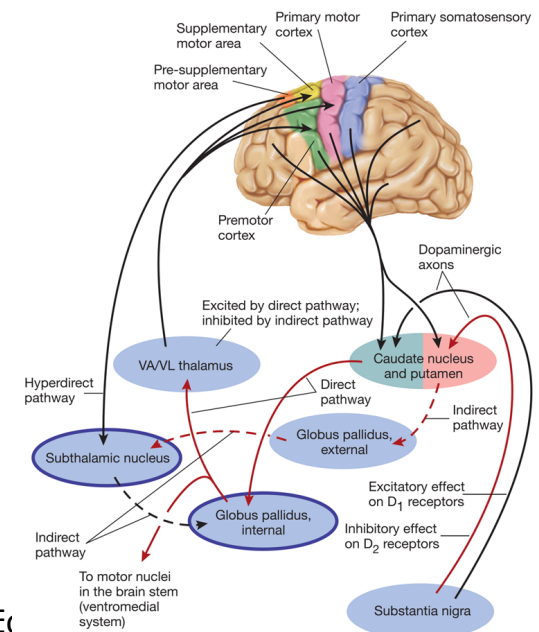
Consequences of Long-term Drug Treatment of Schizophrenia

- Early drugs used for treatment had side effects
 - Parkinson's-like symptoms
 - **Tardive dyskinesia**
 - Involuntary movements of face and neck

The Mesolimbic Dopamine Pathway: Positive Symptoms (2 of 2)

Tardive dyskinesia

- Supersensitivity may occur as a compensatory mechanism in which the receptors are more sensitive when inhibited for a long time
- D₂ receptors inhibited in caudate nucleus and putamen



GO ABOUT YOUR DAY WITH REDUCED INVOLUNTARY MOVEMENTS¹

Once-daily AUSTEDO XR contains the same active ingredient as twice-daily AUSTEDO[®] (deutetrabenazine) tablets. Data on this page is based on twice-daily dosing.¹

In a 12-week clinical study, some people taking AUSTEDO saw significant symptom improvement.¹

The Mesocortical Dopamine Pathway: Negative and Cognitive Symptoms (1 of 3)

Hypofrontality

- Believed to be responsible for negative symptoms
- May also be responsible for hyperactivity in the mesolimbic pathways related to positive symptoms

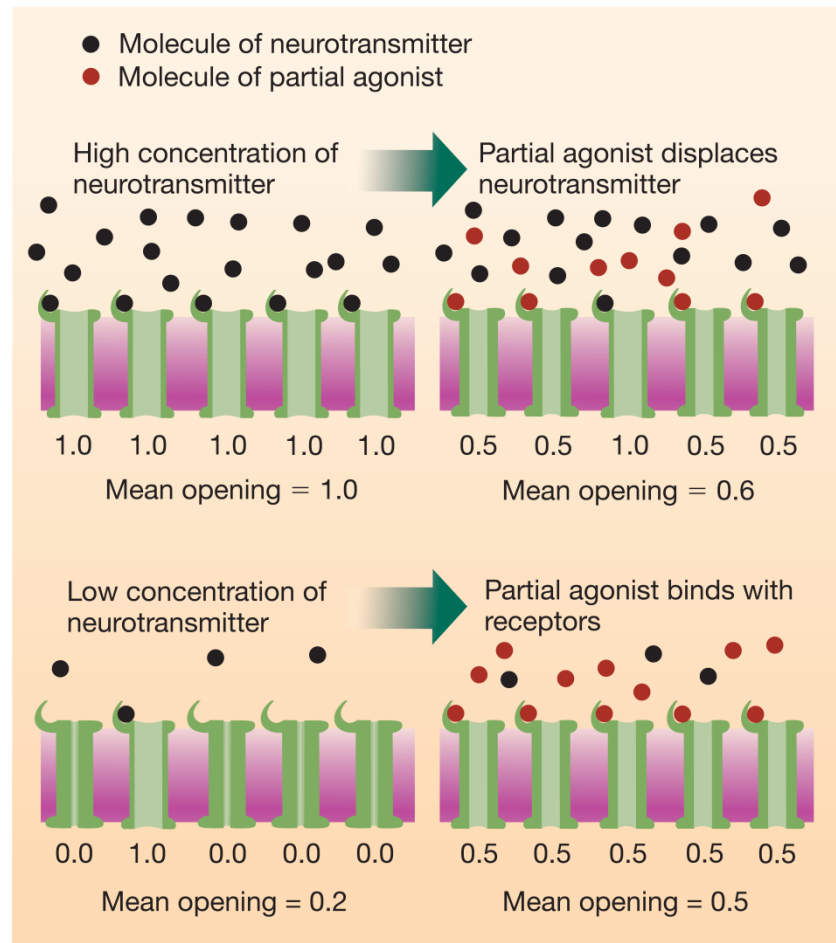
The Mesocortical Dopamine Pathway: Negative and Cognitive Symptoms (2 of 3)

Atypical Antipsychotics

- Unlike original antipsychotics, they increase dopaminergic activity in prefrontal cortex and reduce it in the mesolimbic system
 - Aripiprazole; acts as a **partial agonist** at dopamine receptors
 - Increase dopaminergic activity in the prefrontal cortex
 - Where dopamine is low
 - Decrease dopaminergic activity in the mesolimbic system
 - Where dopamine is high

Figure 16.4

Effects of a Partial Agonist



The diagram explains the differential effects of a partial agonist in regions of high and low concentrations of the normal ligand. Numbers beneath each receptor indicate the degree of opening of the ion channel: 1.0 = fully open, 0.5 = partially open, 0.0 = fully closed. Partial agonists decrease the mean opening when the extracellular concentration of the neurotransmitter is high and increase it when the extracellular concentration of the neurotransmitter is low.

The Mesocortical Dopamine Pathway: Negative and Cognitive Symptoms (2 of 3)

The Role of Glutamate

- Decreased glutamate activity (resulting in hypofrontality) may contribute to negative and positive symptoms
- Chronic low doses of glutamate antagonist drugs produces negative and cognitive symptoms
- PCP and ketamine
 - Indirect antagonist of NMDA receptors

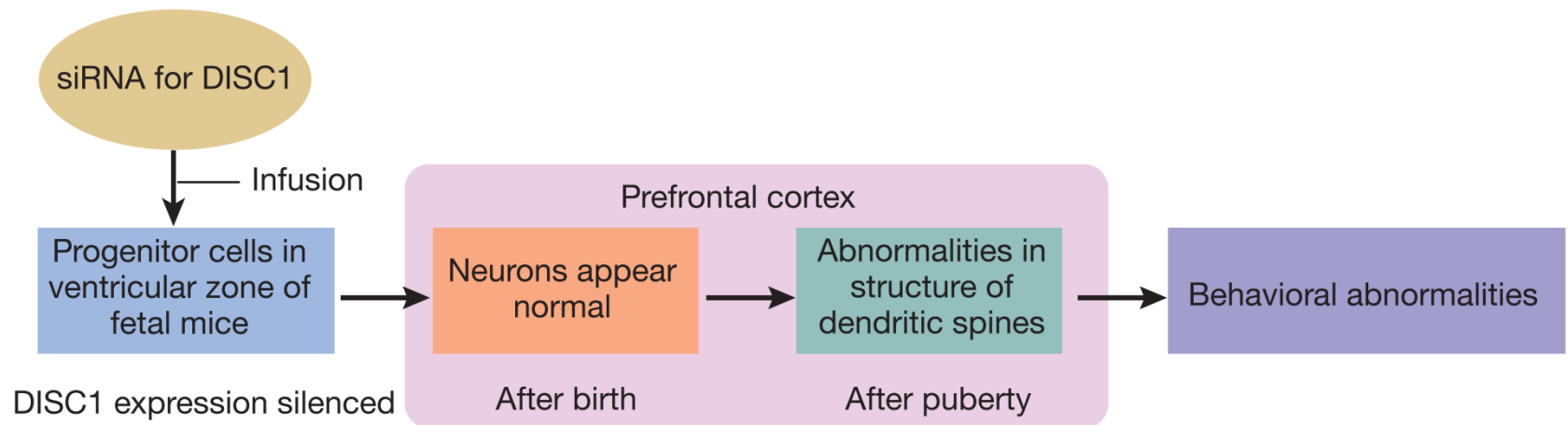
The Mesocortical Dopamine Pathway: Negative and Cognitive Symptoms (3 of 3)

Developmental Changes

- Two theories:
 - Abnormalities in pyramidal neurons of the prefrontal cortex are primary cause of process that leads to schizophrenia
 - Abnormalities in the striatal dopaminergic system are primary cause of process that leads to schizophrenia

Figure 16.5

Role of DISC1 in the Development of Schizophrenia



Niwa et al. (2010) infused an siRNA that prevented the expression of DISC1 in progenitor cells in the ventricular zone of fetal mice. Although neurons in the prefrontal cortex appeared normal after birth, they developed abnormalities in dendritic spines of these neurons after puberty that led to behavioral abnormalities resembling those of schizophrenia. In humans, the symptoms of schizophrenia usually emerge after puberty.

REVIEW

The impact of *Disrupted-in-Schizophrenia 1* (*DISC1*) on the dopaminergic system: a systematic review

T Dahoun^{1,2,3}, SV Trossbach⁴, NJ Brandon⁵, C Korth⁴ and OD Howes^{1,2,3,6}

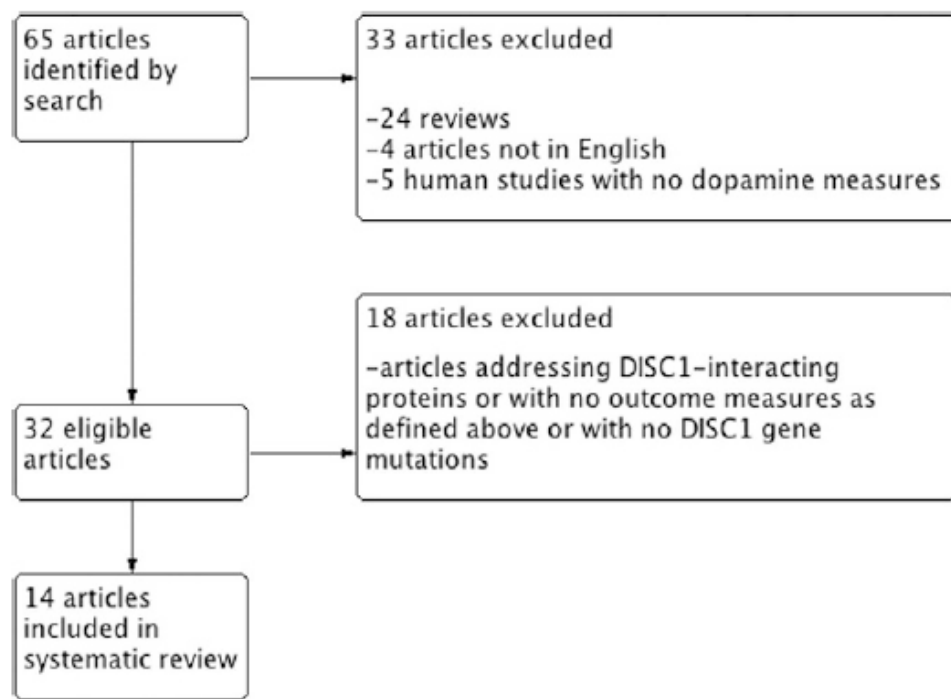


Figure 1. Flow chart of identification, exclusion and inclusion of eligible studies. DISC1, disrupted-in-schizophrenia 1.

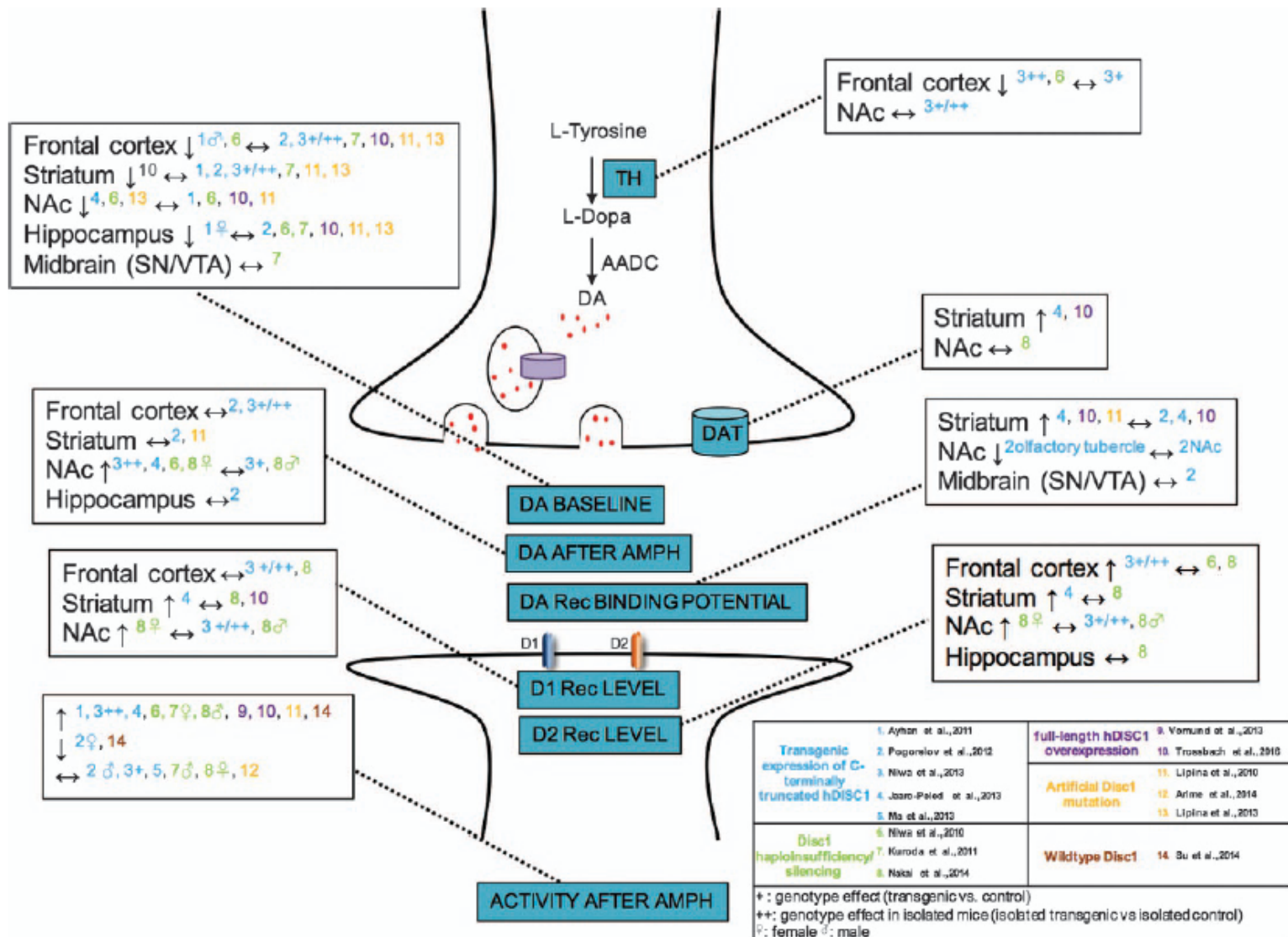
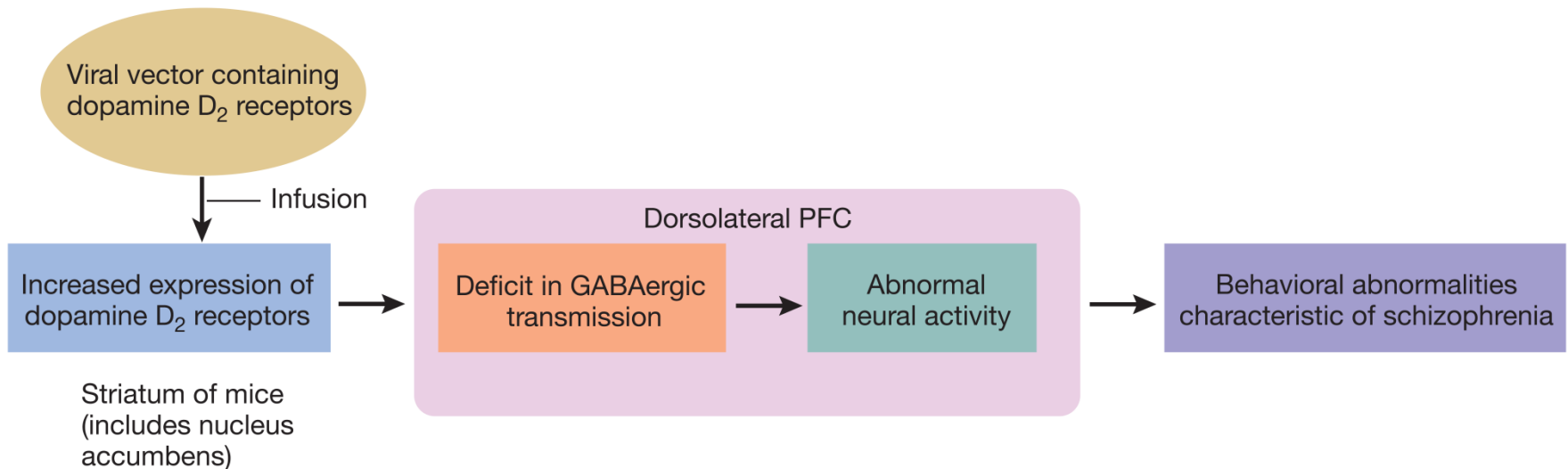


Figure 16.6

Role of Dopamine D2 Receptors in the Development of Schizophrenia



Li et al. (2011) used a viral vector to increase expression of dopamine D2 receptors in the striatum of mice. As a result, GABAergic transmission decreased in the dorsolateral prefrontal cortex, and the abnormal neural activity there led to behavioral deficits characteristic of schizophrenia.

Affective Disorders

- **Description**
- **Genetic Factors**
- **Biological Treatments**
- **Role of the Frontal Cortex**
- **The Monoamine Hypothesis**
- **Role of the 5-HT Transporter**
- **Role of Neurogenesis**
- **Role of Circadian Rhythms**

Description

Learning Objective 16.6 Contrast the symptoms of major depressive disorder and bipolar disorder.

Affective Disorders-What are the differences?

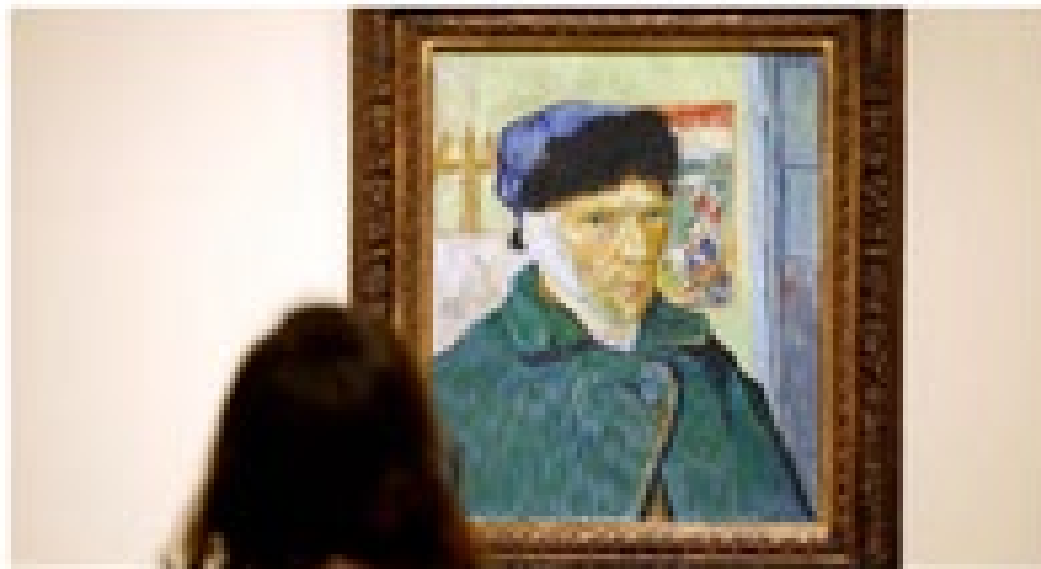
- Characterized by disordered feelings
- Two main types:
 - **Bipolar disorder**
 - **Major depressive disorder (MDD)**

Exploring Vincent van Gogh's Mental Health: Schizophrenia, Bipolar Disorder, or Another Condition?

[Van Gogh's life and mental health](#) | [Possible symptoms](#) | [Possible conditions](#) |

[Van Gogh syndrome](#) | [Recap](#)

More than a century after his passing, researchers are still trying to unravel the mystery of the legendary painter's mental health.



Paul Gauguin



Genetic Factors

Learning Objective 16.7 Describe evidence that supports a genetic contribution to the development of affective disorders.

Factors

- Heritability studies suggest that genetic anomalies are at least partly responsible for affective disorders
 - Close relatives of those with affective disorders are 10 times more likely to develop them
 - Concordance rates: 69 percent in monozygotic twins, versus 13 percent in dizygotic twins

Suicide

15.9 percent in those with major depressive disorder attempt suicide

29.2 percent of people with bipolar attempt suicide

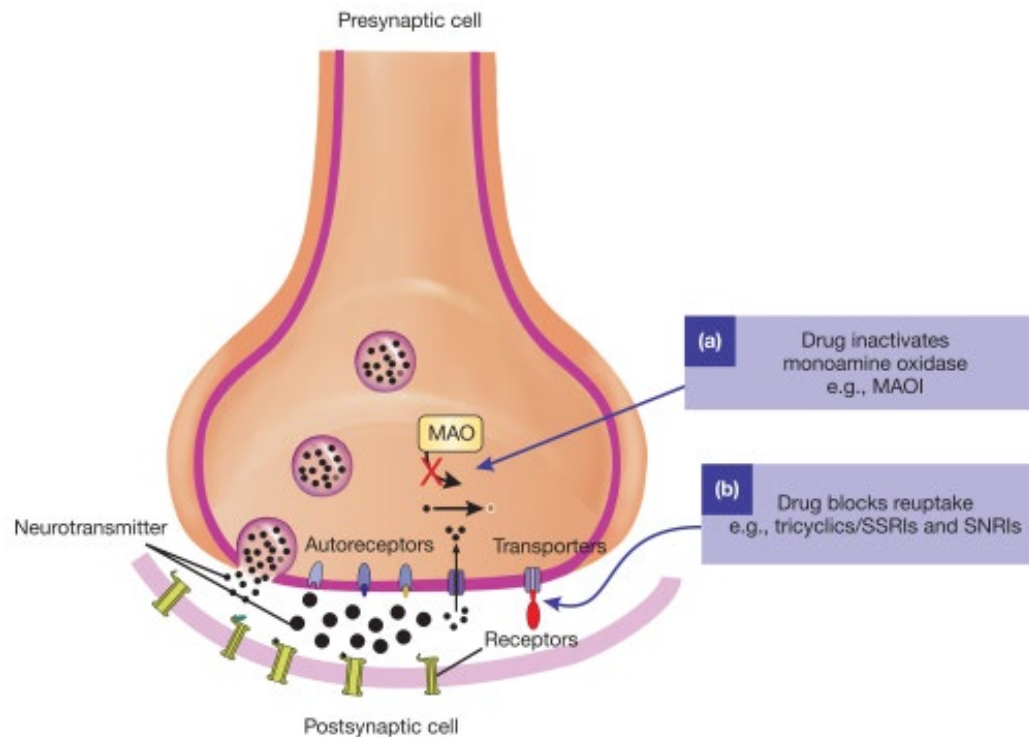
The rate is 28.8 times higher than population

Biological Treatments (1 of 3)

Learning Objective 16.8 List the types of biological treatments for affective disorders.

Pharmacological Treatments

- Antidepressants
 - MAO inhibitors: serotonin and the catecholamines dopamine, adrenaline, and noradrenaline



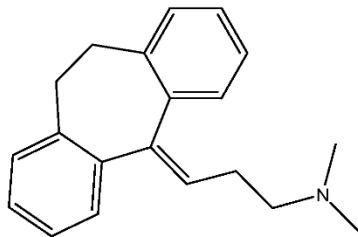
Biological Treatments (1 of 3)

Learning Objective 16.8 List the types of biological treatments for affective disorders.

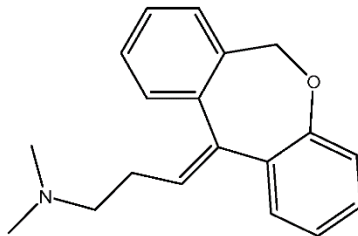
Pharmacological Treatments

- Antidepressants
 - **Tricyclic antidepressants**
 - Inhibit reuptake of 5-HT and norepinephrine

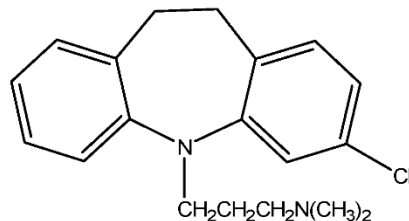
Tertiary TCAs



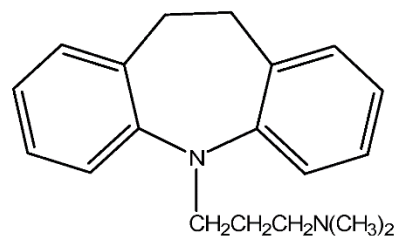
Amitriptyline



Doxepin

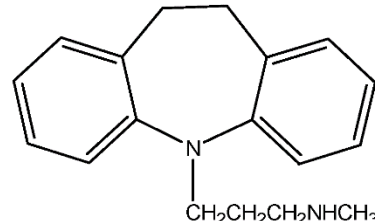


Clomipramine

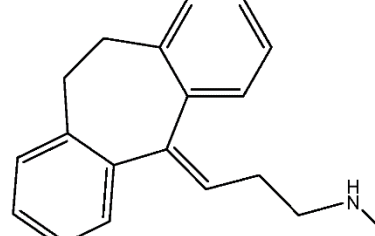


Imipramine

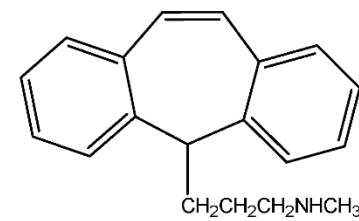
Secondary TCAs



Desipramine



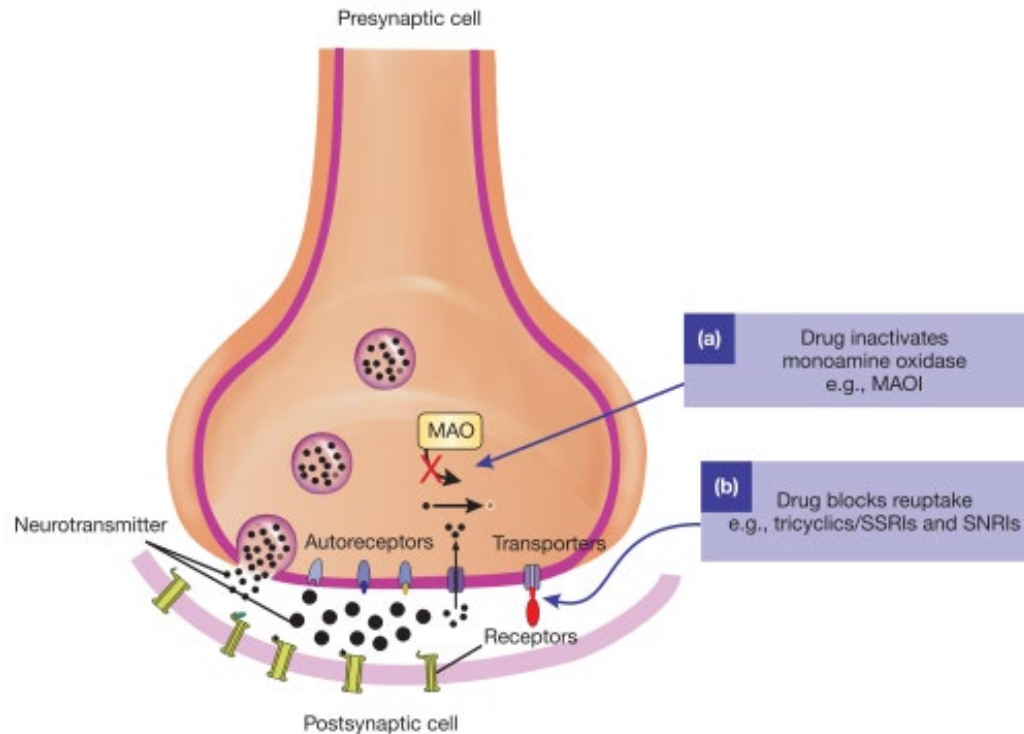
Nortriptyline



Protriptyline

Figure 16.7

Pharmacological Treatment Strategies for Depression



Many pharmacological treatment strategies for depression enhance monoamine neurotransmission. (a) Monoamine oxidase inhibitors prevent the deactivation of monoamines. (b) SSRIs and SNRIs prevent the reuptake of monoamines.

Biological Treatments (1 of 3)

Learning Objective 16.8 List the types of biological treatments for affective disorders.

Pharmacological Treatments

- Antidepressants
 - **Selective serotonin reuptake inhibitors (SSRIs)**
 - Prozac, Celexa, and Paxil
 - **Serotonin and norepinephrine reuptake inhibitors (SNRIs)**
 - Antagonist action on noradrenergic transporters

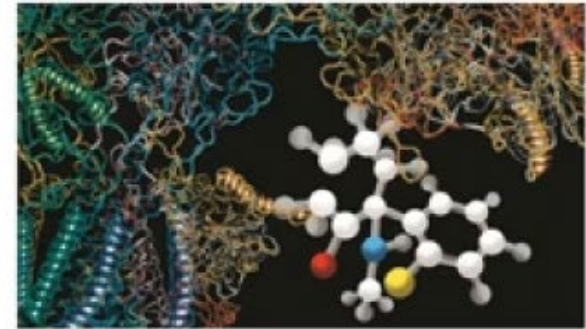
Biological Treatments (1 of 3)

Learning Objective 16.8 List the types of biological treatments for affective disorders.

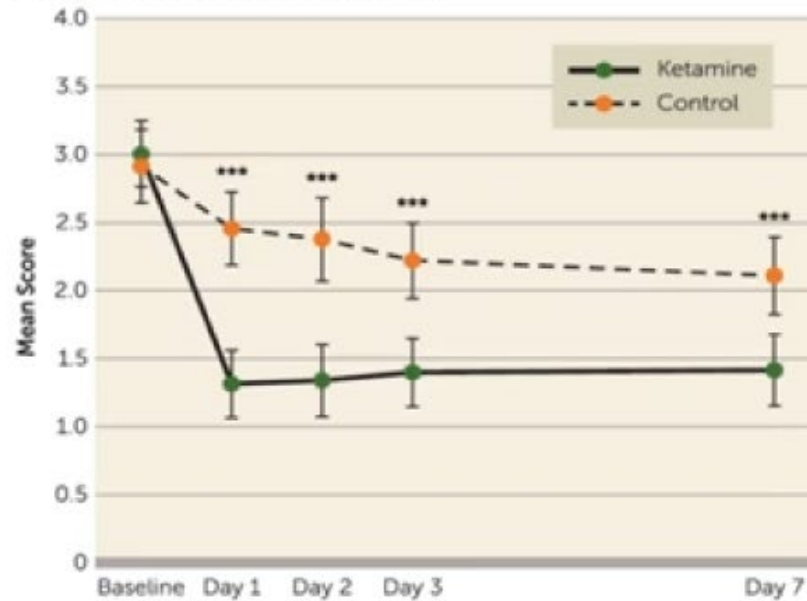
Pharmacological Treatments

- Ketamine
 - For treatment resistant depression
- Lithium
 - For bipolar
 - 70-80% of those with bipolar show positive response to lithium within 1-2 weeks
 - Side effects: hand tremors, weight gain, excessive urine production, and thirst

Ketamine is also anti-suicidal in adults, even after controlling for its antidepressant effects



A. Clinician-Reported Suicidal Ideation



Biological Treatments (2 of 3)

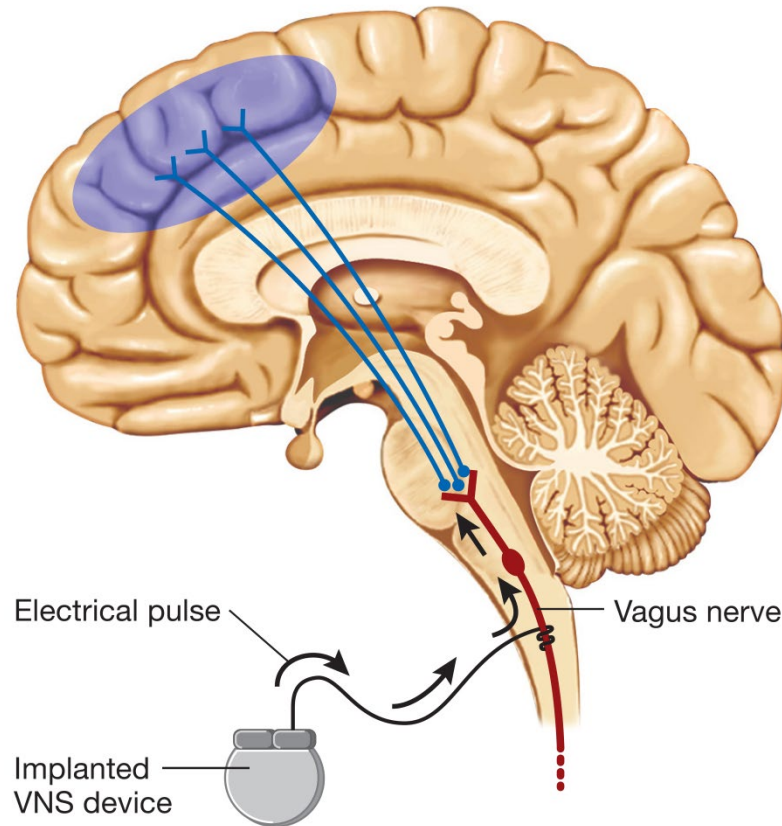
Electroconvulsive Therapy

- ECT involves brief electrical shock used to induce a seizure; used to treat mania and depression



Figure 16.8

Vagus Nerve Stimulation



Electrical stimulation of the vagus nerve activates several regions of the brain. VNS = Vagus nerve stimulation. Source: (Stahl, S. M. (2008). Antipsychotics and mood stabilizers: Stahl's essential psychopharmacology. Cambridge University Press.)

Biological Treatments (3 of 3)

Transcranial Magnetic Stimulation

- TMS provides similar benefits to ECT without risk of cognitive impairments or memory loss
- Used to treat depression

Deep Brain Stimulation

- DBS also useful for treatment-resistant depression

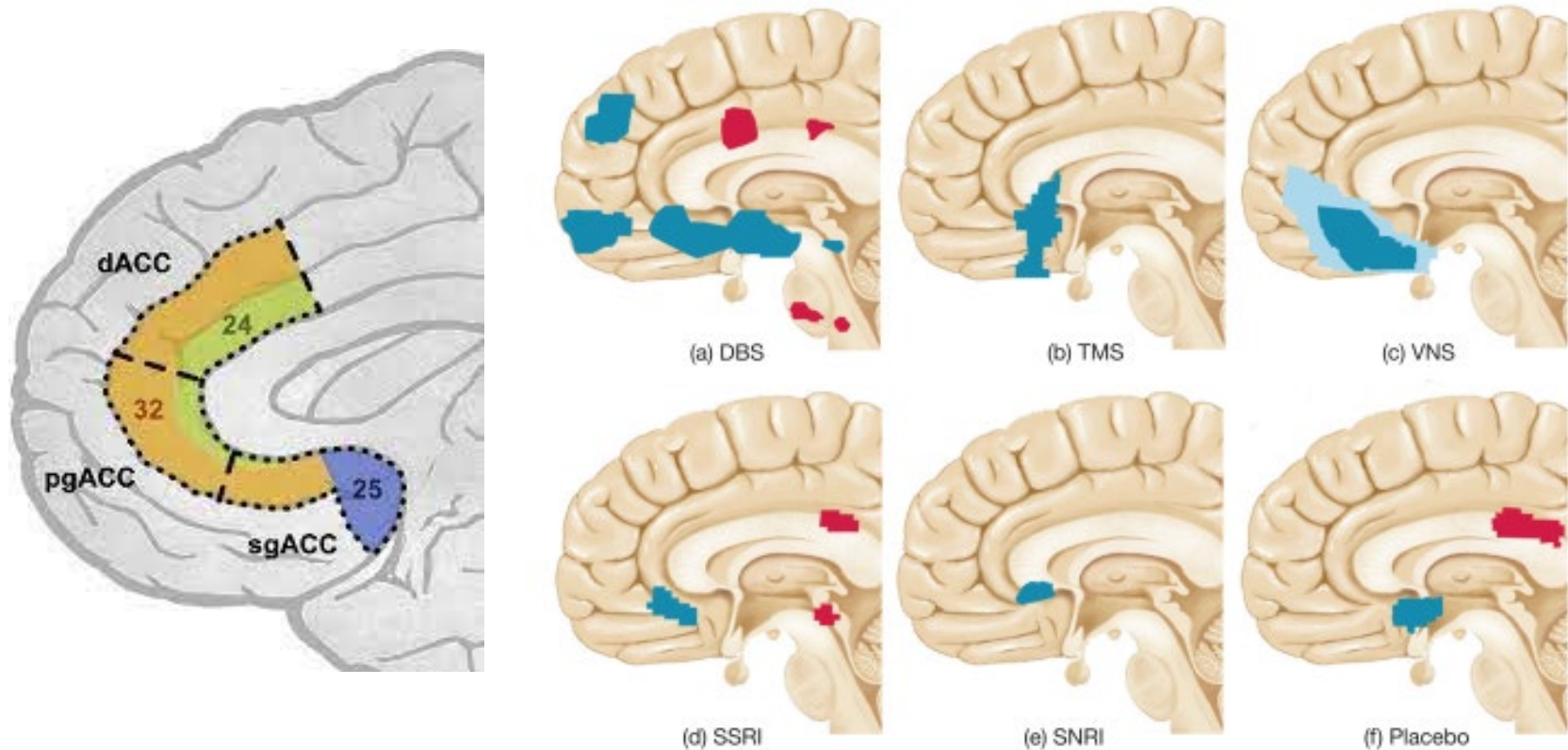
Role of the Frontal Cortex

Learning Objective 16.9 Describe the role of the frontal cortex in depression.

Frontal Cortex

- Depressed patients show hyperactivity of subgenual ACC along with decreased activity in other regions of frontal cortex
 - Dorsolateral PFC, the ventrolateral PFC, the ventromedial PFC, and the orbitofrontal cortex

Figure 16.9 Decreased Activation of the Subgenual ACC After a Variety of Successful Treatments for Depression



The figure shows a standard drawing of an anterior midsagittal view of the human brain with tracings of regions of increased (red) or decreased (blue) activation seen in functional imaging studies of brain responses to successful treatment for the symptoms of depression. al. (2002), (e) Kennedy et al. (2007), (f) Mayberg et al. (2002).)

The Monoamine Hypothesis

Learning Objective 16.10 Summarize the evidence in support of the monoamine theory of affective disorders.

Monoamine Hypothesis

- Depression is caused by insufficient activity of monoaminergic neurons
- Research focuses on role of norepinephrine and serotonin
 - Depletion of tryptophan in brain causes recurrence of depressive symptoms
 - 5-HT plays a role in mood, although exact nature is unknown

Role of the 5-HT Transporter

Learning Objective 16.11 Critique the evidence for the role of the promoter region of the 5-HT transporter gene in affective disorders.

Role of Serotonin Transporter

- Some studies found that stressful life events increase probability of depression for those with one or two copies of the short alleles for the 5-HTT promoter
- However, role in depression is unclear and needs further research

How the serotonin transporter 5-HTTLPR polymorphism influences amygdala function: the roles of *in vivo* serotonin transporter expression and amygdala structure

A Kobiella^{1,2}, M Reimold³, DE Ulshöfer⁴, VN Ikonomidou⁵, C Vollmert², S Vollstädt-Klein², M Rietschel⁶, G Reischl⁷, A Heinz⁸ and MN Smolka^{1,9}

The serotonin transporter-linked promoter region (5-HTTLPR) polymorphism of the serotonin transporter gene is associated with amygdala response during negative emotion. The aim of this study was to investigate whether this genotype effect on amygdala function is mediated by current serotonin transporter (5-HTT) levels or rather by genetically induced influences during neurodevelopment, shaping brain structure. A total of 54 healthy subjects underwent functional and structural magnetic resonance imaging, [¹¹C]DASB positron emission tomography and 5-HTTLPR genotyping to analyze the interrelationships between amygdala activation during processing of unpleasant stimuli, 5-HTTLPR genotype, amygdala volumes and 5-HTT levels in the midbrain and in other brain regions. In line with previous research, carriers of the short allele (S) showed increased amygdala activation. Path analysis demonstrated that this genotype effect was not procured by current 5-HTT availability but by amygdala structure, with smaller amygdala volumes in the S than in the LL genotype, as well as smaller volumes being associated with increased amygdala activation. Our findings stress the role of genetic effects during neurodevelopment.

Translational Psychiatry (2011) 1, e37; doi:10.1038/tp.2011.29; published online 30 August 2011

Environment

(e.g. seasonal effects, stress, smoking)



5-HTT Expression



5-HTTLPR

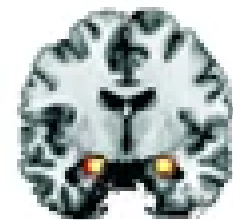


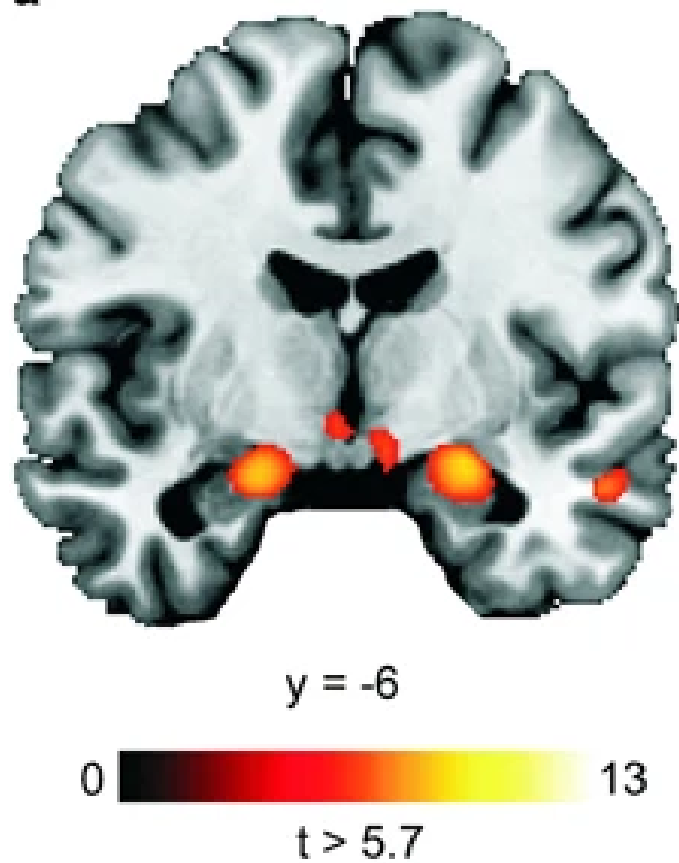
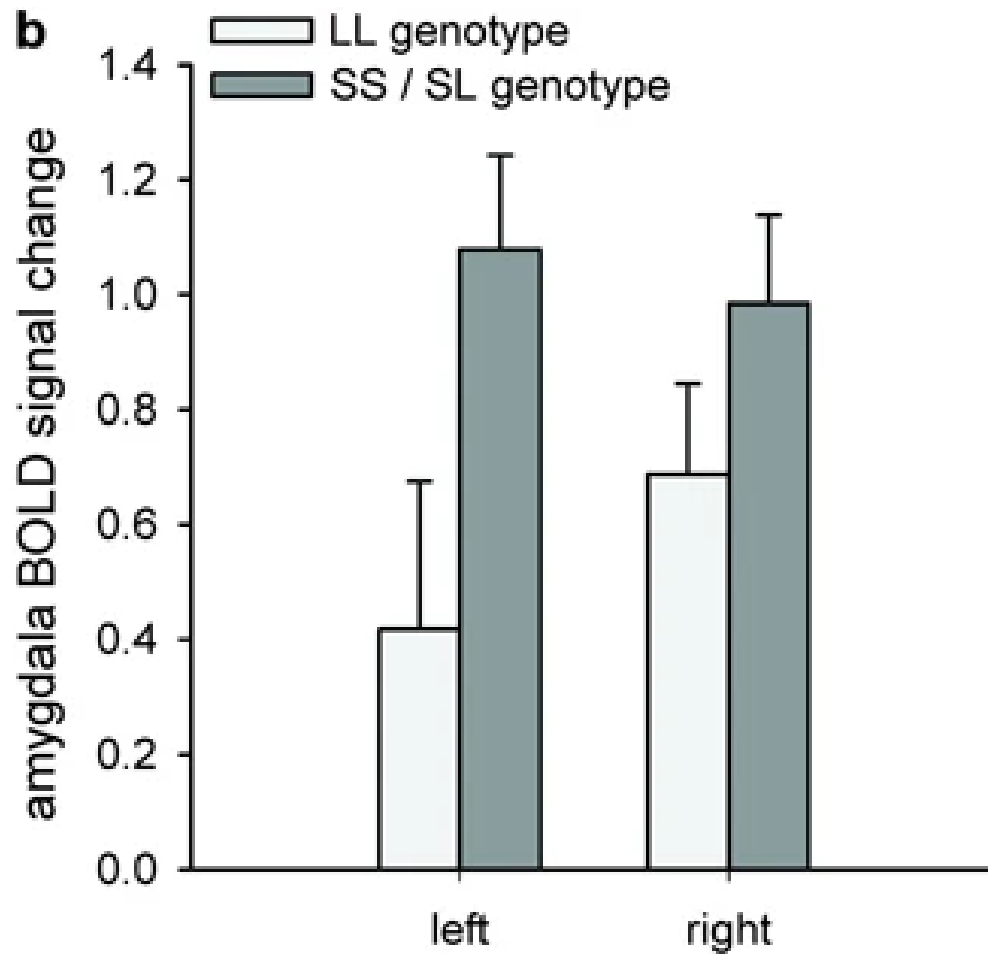
Neurodevelopment

(e.g. effects on brain structure and function)



Amygdala Reactivity



a**b**

Role of Neurogenesis

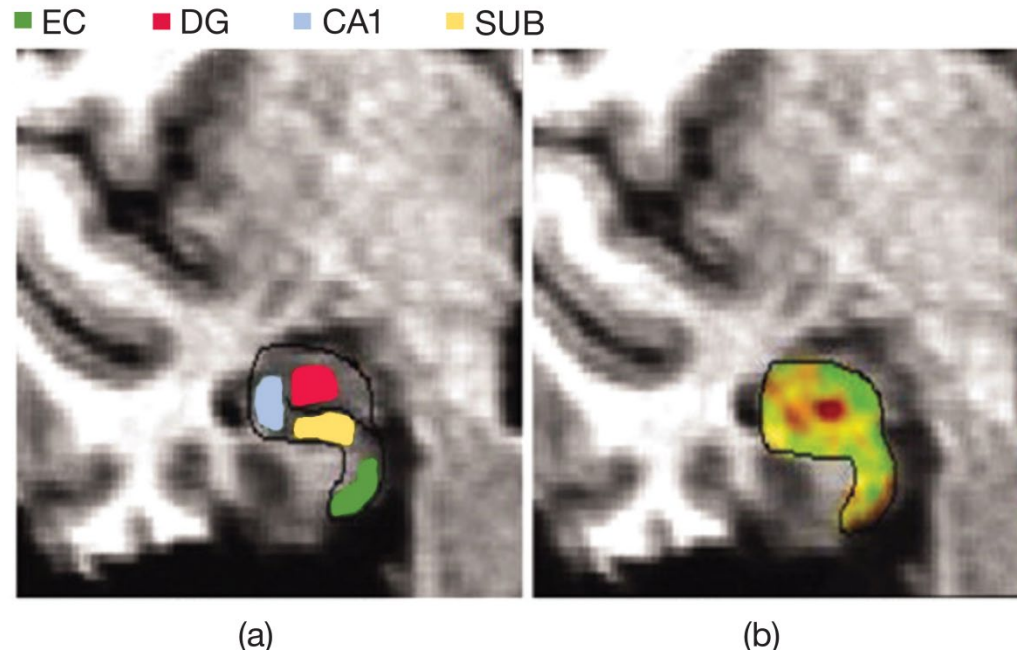
Learning Objective 16.12 Explain the relationship between affective disorders, their treatment, and neurogenesis.

Neurogenesis and Depression

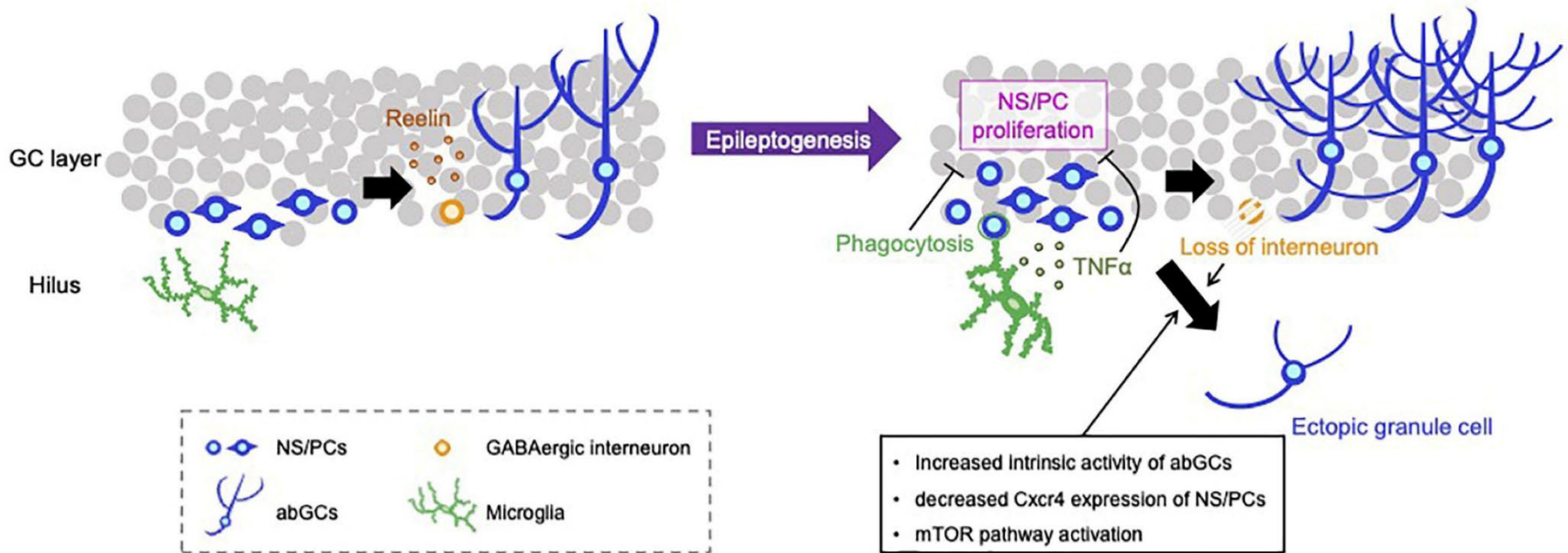
- Stress and depression are associated with reduced hippocampal neurogenesis
- Antidepressant treatment increases hippocampal neurogenesis in lab animals
- Evidence that exercise induces neurogenesis in the human brain

Figure 16.10

Exercise and Neurogenesis



The scans show the effect of a program of aerobic exercise on the blood volume of regions of the human hippocampal formation. This is an indirect measure of neurogenesis. (a) Subregions of the hippocampus. EC = entorhinal cortex, DG = dentate gyrus, SUB = subiculum. (b) Regional blood volume. “Hotter” colors indicate increased blood volume. (Source: From Pereira, A. C., Huddleston, D. E., Brickman, A. M., et al., An in vivo correlate of exercise-induced neurogenesis in the adult dentate gyrus, *Proceedings of the National Academy of Sciences, USA* 2007, 104 5638–5643. Reprinted with permission.)



Role of Circadian Rhythms

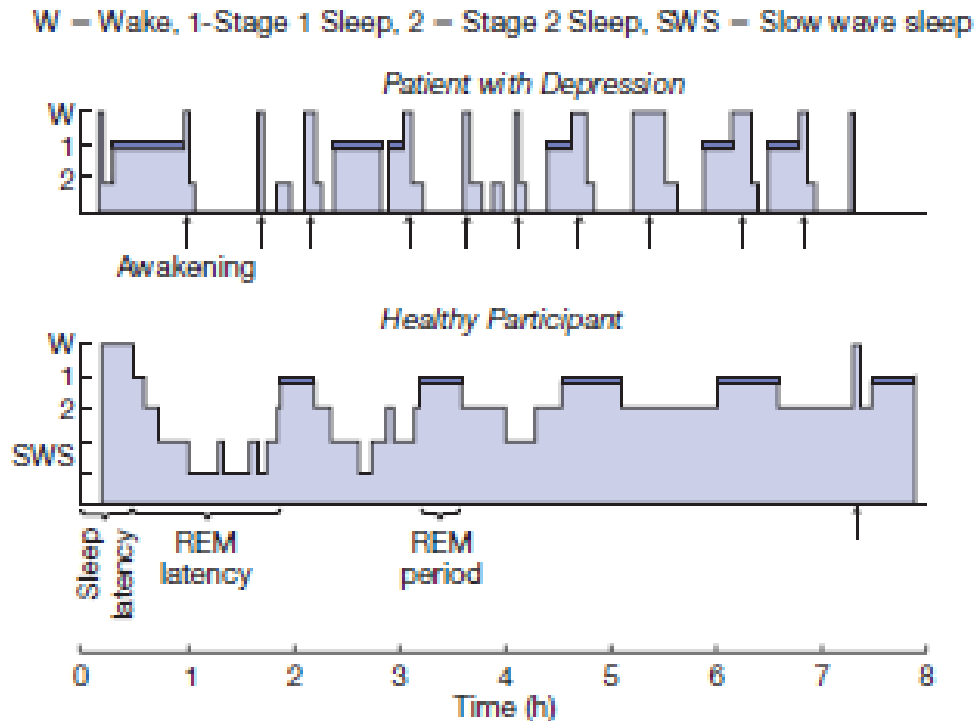
Learning Objective 16.13 Summarize the involvement of circadian rhythms in affective disorders.

Circadian Rhythms and Depression

- Sleep disturbances are characteristic of affective disorders
- Sleep deprivation can reduce depression in some people
- **Seasonal affective disorder** is a unique type of depression
 - Can be treated by **phototherapy**

Figure 16.11

Sleep and Depression



The diagram illustrates patterns of the stages of sleep of a healthy participant and of a patient with major depression. Note the reduced sleep latency, reduced REM latency, lack of slow-wave sleep, and general fragmentation of sleep (arrows) in the patient with depression. (Source: Based on Gillin, J. C., and Borbély, A. A., Sleep: A neurobiological window on affective disorders, Trends in Neurosciences, 1985, 8, 537–542.)

Emotional Wellness Toolkit

- <https://www.nih.gov/health-information/emotional-wellness-toolkit>

Anxiety Disorders

- **Symptoms**
- **Genetic and Environmental Factors**
- **Treatment**

Symptoms: Anxiety

Learning Objective 16.14 List the symptoms of anxiety disorders.

Anxiety Disorders

- Characterized by unrealistic, unfounded fear and anxiety; most common psychiatric disorder
- With a lifetime prevalence of approximately 28 percent, anxiety disorders are the most common psychiatric disorders.

Symptoms: Anxiety

Learning Objective 16.14 List the symptoms of anxiety disorders.

Anxiety Disorders

What are some of the characteristics of anxiety disorders and what are some types?

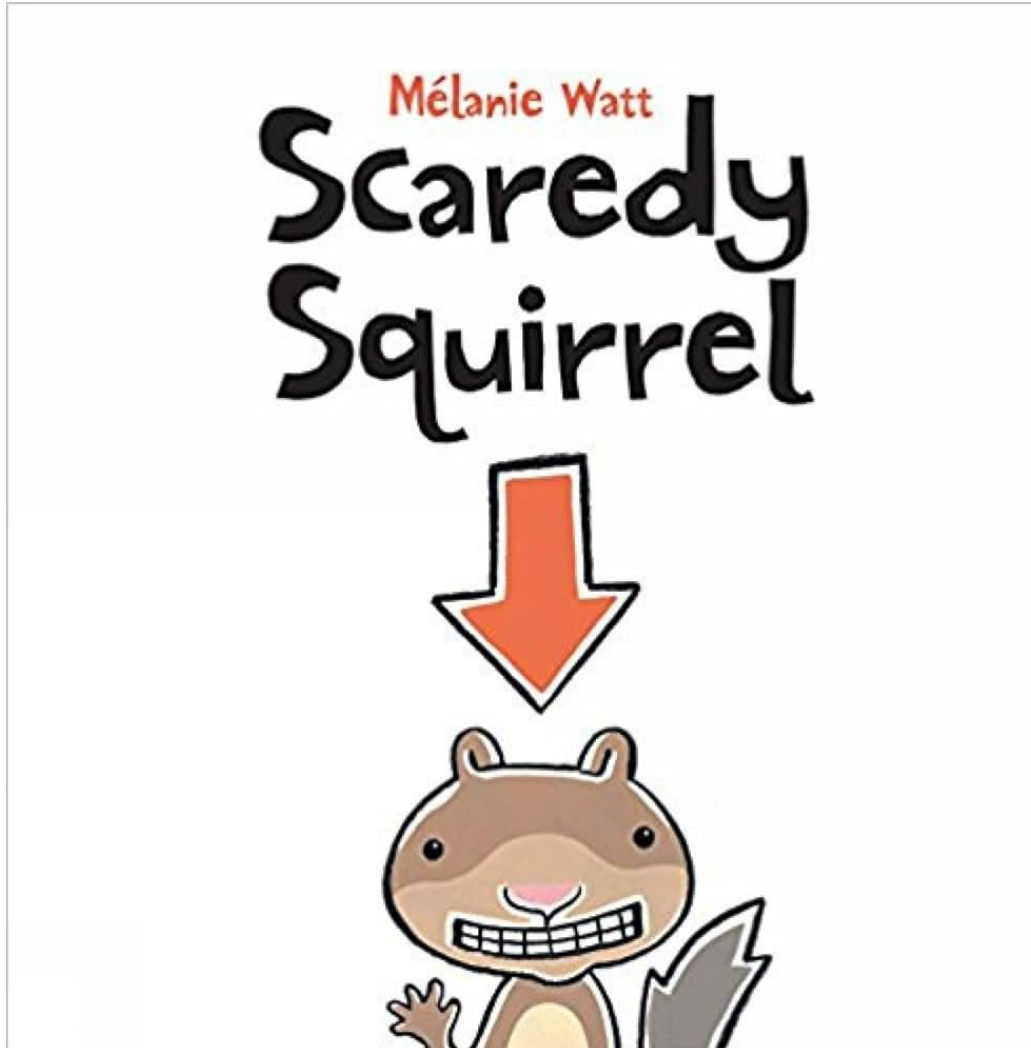
Symptoms: Anxiety

Learning Objective 16.14 List the symptoms of anxiety disorders.

Anxiety Disorders

- Characterized by unrealistic, unfounded fear and anxiety; most common psychiatric disorder
 - **Panic disorder**
 - **Anticipatory anxiety**
 - **Agoraphobia**
 - **Generalized anxiety disorder**
 - **Social anxiety disorder**

Agoraphobia





Rumination: A Cycle of Negative Thinking

March 05, 2020 Anxiety, Depression, Patients and Families

Rumination involves repetitive thinking or dwelling on negative feelings and distress and their causes and consequences. The repetitive, negative aspect of rumination can contribute to the development of depression or anxiety and can worsen existing conditions.

When a person who is in a depressed mood ruminates, they are more likely to "remember more negative things that happened to them in the past, they interpret situations in their current lives more negatively, and they are more hopeless about the future."¹ The preoccupation with problems also makes it difficult to move beyond to allow for a focus on problem solving. Even in people without depression or anxiety, rumination can contribute to negative emotions. This can become a cycle where the more a person ruminates, the worse they feel, which then contributes to more rumination.²

A study in the University of Liverpool in the United Kingdom looked at the connections between a person's circumstances and past experiences and development of depression and anxiety. Researchers, led by Peter Kinderman, Ph.D., found that the most important way that a person's past experiences, such as traumatic life events, led to depression or anxiety was "by leading a person to ruminate and blame themselves for the problem."³ "Depression and anxiety are not simple conditions and there is no single cause," Kinderman

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[Depression \(78\)](#)

[Diverse populations \(12\)](#)

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Genetic and Environmental Factors: Anxiety

Amygdala and vmPFC

- Adolescents with generalized anxiety disorder show increased activation of amygdala and decreased activation of vmPFC
 - vmPFC-role in extinction and inhibition of fear and anxiety

Treatment: Anxiety

Learning Objective 16.16 Summarize pharmacological treatments for anxiety disorders.

Pharmacological Treatment

- Benzodiazepines
 - Fast acting; often used for emergency treatment
 - GABA_A receptor agonist
 - Less appropriate for long-term treatment
- SSRIs
 - Preferred treatment option
 - Used in conjunction with cognitive behavior therapy

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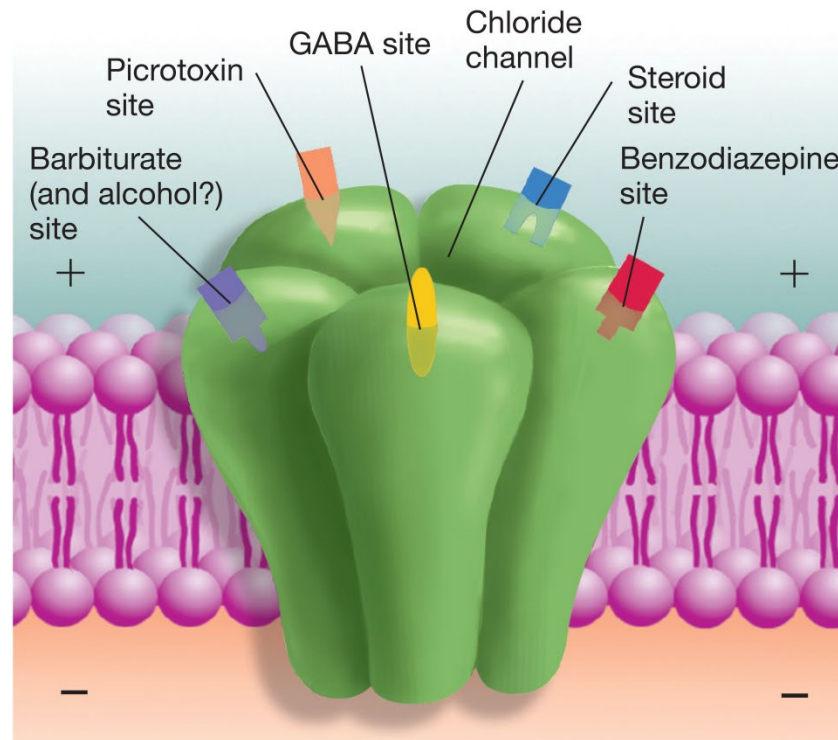
HBO ORIGINAL
THE
WHITE LOTUS

NEW SEASON

FEB 16 | max

Figure 16.12

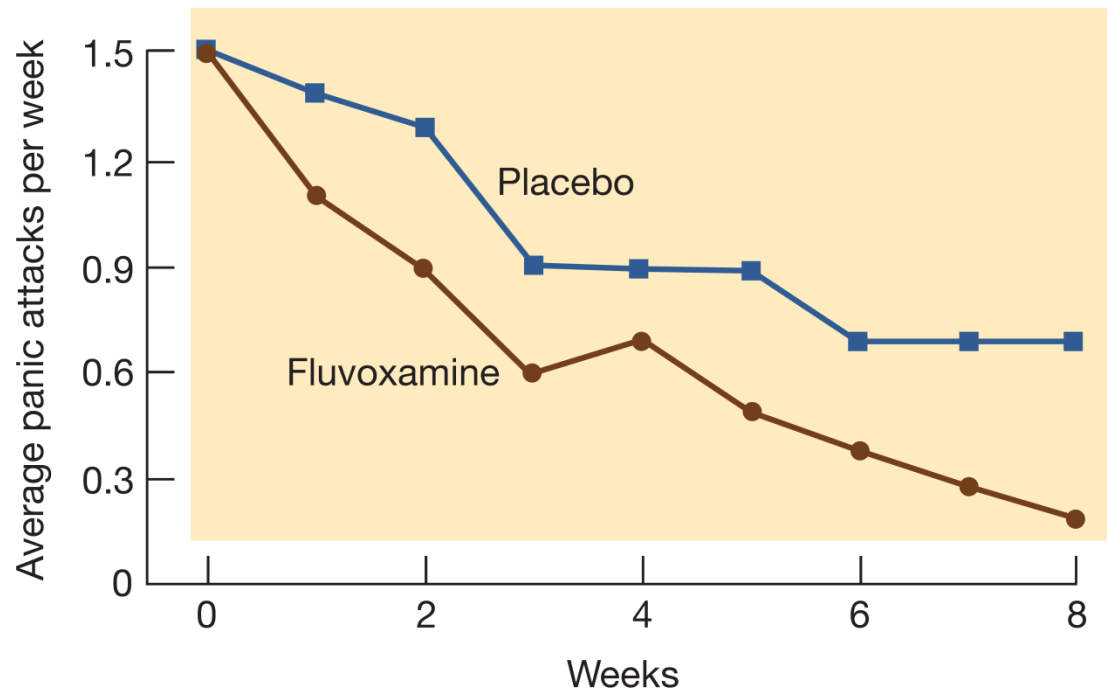
Benzodiazepines Bind to GABAA Receptors



The GABAA receptor has several binding sites, including one which binds with benzodiazepines.

Figure 16.13

Fluvoxamine and Panic Disorder



The graph shows the effects of fluvoxamine (an SSRI) on the severity of panic disorder. Source: (Asnis, Gregory M., et al. "Fluvoxamine in the treatment of panic disorder: a multi-center, double-blind, placebo-controlled study in outpatients." *Psychiatry Research* 103.1 (2001): 1–14.)

JAMA Psychiatry | Original Investigation

Mindfulness-Based Stress Reduction vs Escitalopram for the Treatment of Adults With Anxiety Disorders A Randomized Clinical Trial

Elizabeth A. Hoge, MD; Eric Bui, MD, PhD; Mihriye Mete, PhD; Mary Ann Dutton, PhD; Amanda W. Baker, PhD; Naomi M. Simon, MD, MSc

JAMA Psychiatry. doi:10.1001/jamapsychiatry.2022.3679
Published online November 9, 2022.



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This sucks. Though we're not sure what John the Baptist was thinking in this 15th century painting, most people don't like being alone with their thoughts. HIERONYMUS BOSCH, SAINT JOHN THE BAPTIST IN THE WILDERNESS, CA. 1489-1500; MUSEO LÁZARO GALDIANO, MADRID

People would rather be electrically shocked than left alone with their thoughts

By [Nadia Whitehead](#) | Jul. 3, 2014, 2:00 PM

At some point today you will disengage from the rest of the world and just think. It could happen any number of ways: if your mind wanders from work, while you're sitting in traffic, or if you just take a quiet moment to reflect. But as frequently as we drift into our own thoughts, a new study suggests that many of us don't like it. In fact, some people even prefer an electric shock to being left alone with their minds.

"I'm really excited to see this paper," says Matthew Killingsworth, a psychologist at the University of California (UC), San Francisco, who says his own work has turned up a similar result. "When people



Pea



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- 15 minutes
- Subjects previously said that they would pay money than be shocked
- 67% of men and 25% of women chose to inflict it on themselves rather than just sit there quietly and think, the team reports online today in Science.

SOCIAL PSYCHOLOGY

Just think: The challenges of the disengaged mind

Timothy D. Wilson,^{1*} David A. Reinhard,¹ Erin C. Westgate,¹ Daniel T. Gilbert,²
Nicole Ellerbeck,¹ Cheryl Hahn,¹ Casey L. Brown,¹ Adi Shaked¹

In 11 studies, we found that participants typically did not enjoy spending 6 to 15 minutes in a room by themselves with nothing to do but think, that they enjoyed doing mundane external activities much more, and that many preferred to administer electric shocks to themselves instead of being left alone with their thoughts. Most people seem to prefer to be doing something rather than nothing, even if that something is negative.

“The mind is its own place, and in it self/
Can make a Heav’n of Hell, a Hell of Heav’n.”

– John Milton, *Paradise Lost*

unrelated thought, namely cases in which people are trying to attend to an external task (such as reading a book), but their minds wander involuntarily (7, 8). In such cases, people tend to be happier when their minds are engaged in what

“If you knew Time as well as I do,' said the Hatter, 'you wouldn't talk about wasting IT.'”

- Hatter in Alice's Adventures in Wonderland by Lewis Carroll



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Obsessive-Compulsive Disorder

- Symptoms
- Genetic and Environmental Factors
- Treatment



Symptoms: OCD

Learning Objective 16.17 List the symptoms of OCD.

What is Obsessive-Compulsive Disorder?

Symptoms: OCD

Learning Objective 16.17 List the symptoms of OCD.

Obsessive-Compulsive Disorder

- Individuals experience unwanted thoughts and uncontrollable behaviors; symptoms commonly begin in young adulthood
- Four categories: counting, checking, cleaning, and avoidance

Genetic and Environmental Factors: OCD

Learning Objective 16.18 Describe the roles of genetic and environmental factors in the development of OCD.

Genetic Factors

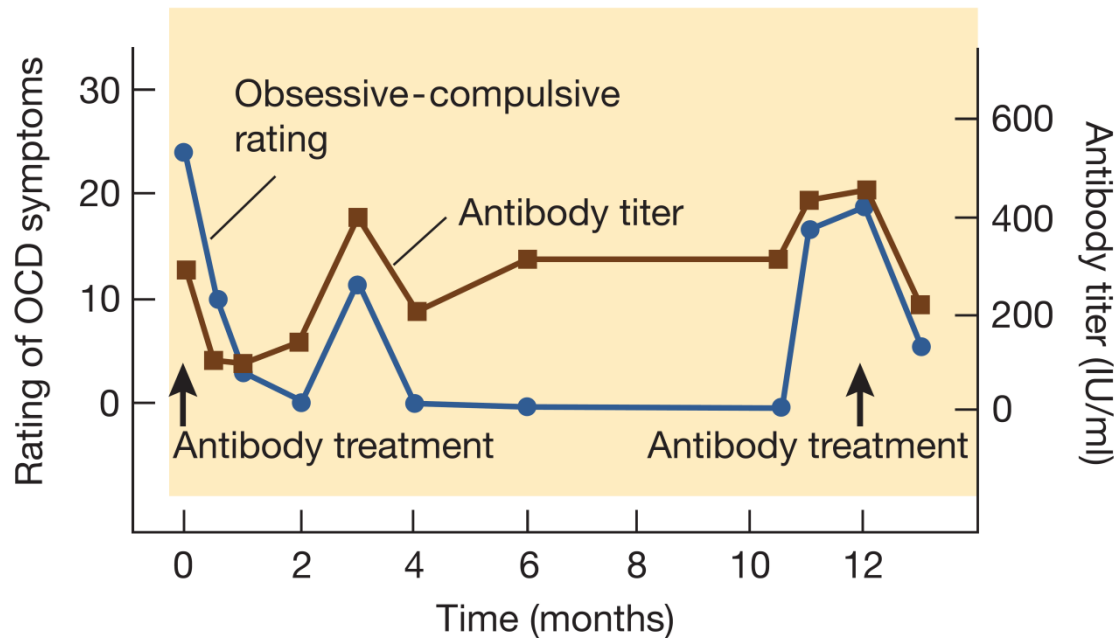
- Heredity plays a role; greater concordance rate in monozygotic twins than in dizygotic twins

Environmental Factors

- Brain damage at birth, encephalitis, and head trauma
- Group A b-hemolytic streptococcal infection
 - Autoimmune disorder

Figure 16.14

OCD and Streptococcal Hemolytic Infection



The graph shows the parallel course of a child's symptoms and the level of antistreptococcal DNA-B in her blood, which indicates the presence of an active infection. This relationship provides evidence that a group A β -hemolytic streptococcal infection can produce the symptoms of OCD, presumably by affecting the basal ganglia. (Based on Perlmutter, S. J., Garvey, M. A., Castellanos, X., Mittleman, B. B., Giedd, J., Rapoport, J. L., & Swedo, S. E. (1998). A case of pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections. *American Journal of Psychiatry*, 155(11), 1592–1598.)

Genetic and Environmental Factors: OCD

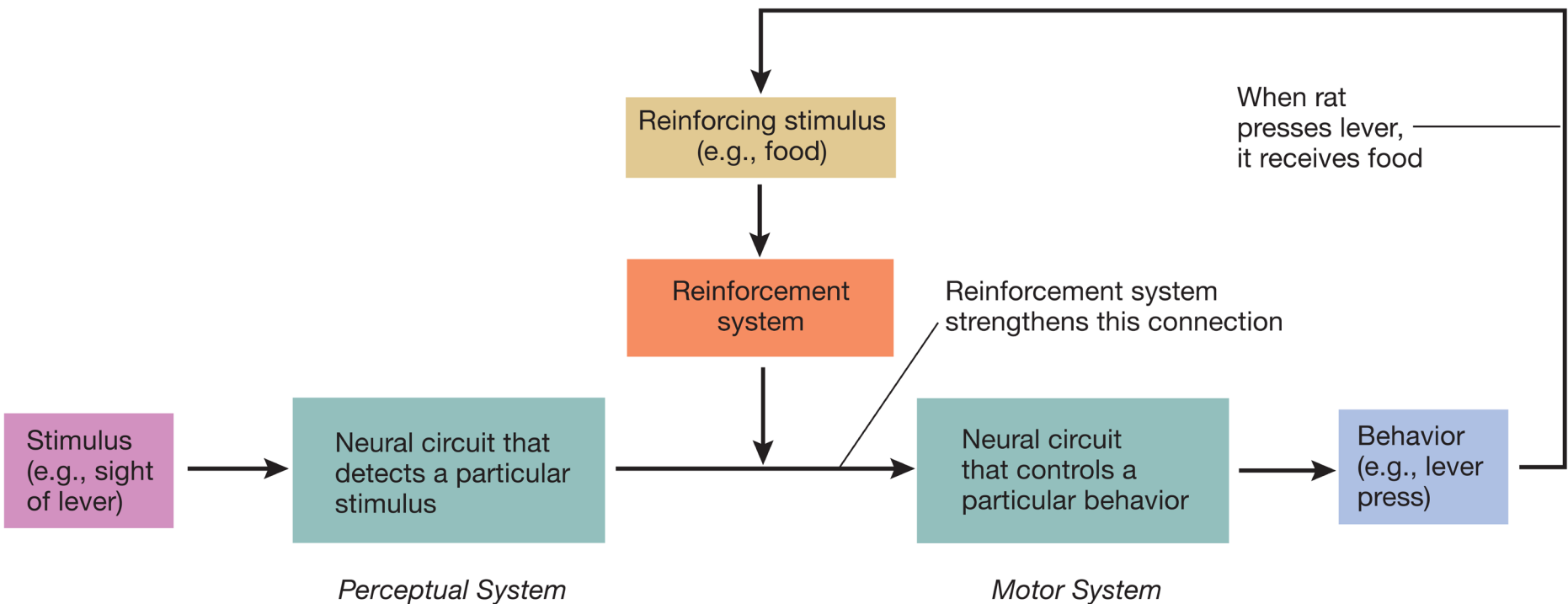
Learning Objective 16.18 Describe the roles of genetic and environmental factors in the development of OCD.

Brain changes

- Damage to the basal ganglia
- There was an increase in size to the basal ganglia for those with beta-hemolytic streptococcus
- Increased activity of frontal lobe and caudate nucleus in those with OCD

Figure 13.4

A Simple Neural Model of Operant Conditioning





Impacts of Stressful Life Events and Traumatic Experiences on Onset of Obsessive-Compulsive Disorder

Keitaro Murayama¹, Tomohiro Nakao^{1}, Aikana Ohno^{1,2}, Sae Tsuruta³, Hirofumi Tomiyama¹, Suguru Hasuzawa¹, Taro Mizobe¹, Kenta Kato¹ and Shigenobu Kanba¹*

Out of 281 patients with OCD

- 172 (61.2%) participants had experienced various stressful life events
- 98 (34%) participants had had traumatic experiences before the onset.

Treatment: OCD

Learning Objective 16.19 Summarize treatments for OCD.

Drug Therapies

- Serotonergic agonists such as clomipramine, fluoxetine, and fluvoxamine used for treating OCD
 - Clomipramine also used for three other compulsions
 - Researchers believe symptoms relieved by increasing the activity of serotonergic pathways with inhibitory role in behaviors

Other Options

- Deep brain stimulation may also be effective

The Nobel Prize in Physiology or Medicine 1949

More ▾

Walter Hess
Egas Moniz

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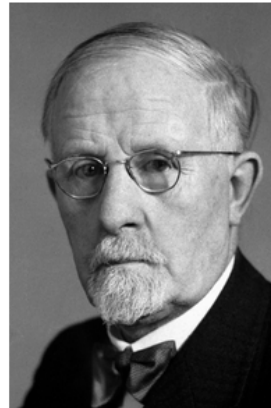


Photo from the Nobel Foundation archive.

Walter Rudolf Hess

Prize share: 1/2



Photo from the Nobel Foundation archive.

**Antonio Caetano de
Abreu Freire Egas
Moniz**

Prize share: 1/2

The Nobel Prize in Physiology or Medicine 1949 was divided equally between Walter Rudolf Hess "for his discovery of the functional organization of the interbrain as a coordinator of the activities of the internal organs" and Antonio Caetano de Abreu Freire Egas Moniz "for his discovery of the therapeutic value of leucotomy in certain psychoses."

To cite this section

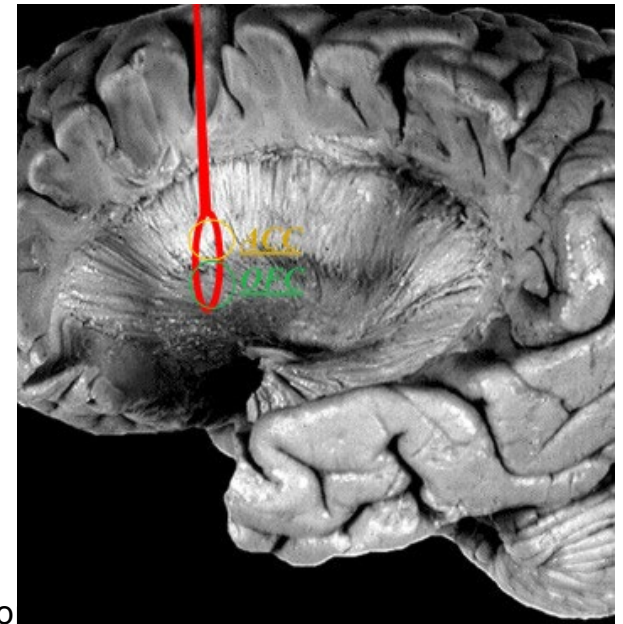
MLA style: The Nobel Prize in Physiology or Medicine 1949. NobelPrize.org. Nobel Media AB 2019. Tue. 23 Apr 2019.

<<https://www.nobelprize.org/prizes/medicine/1949/summary/>>

Treatment: OCD

Surgical Treatment

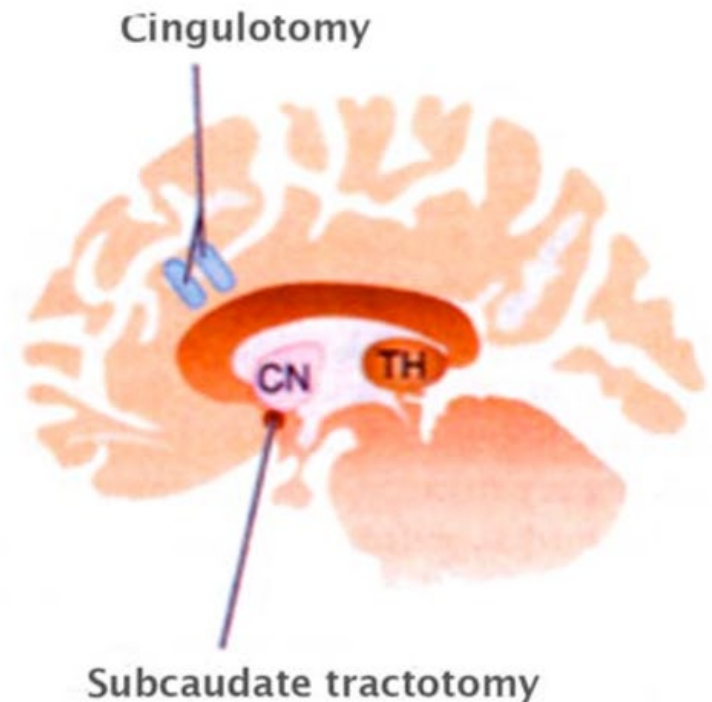
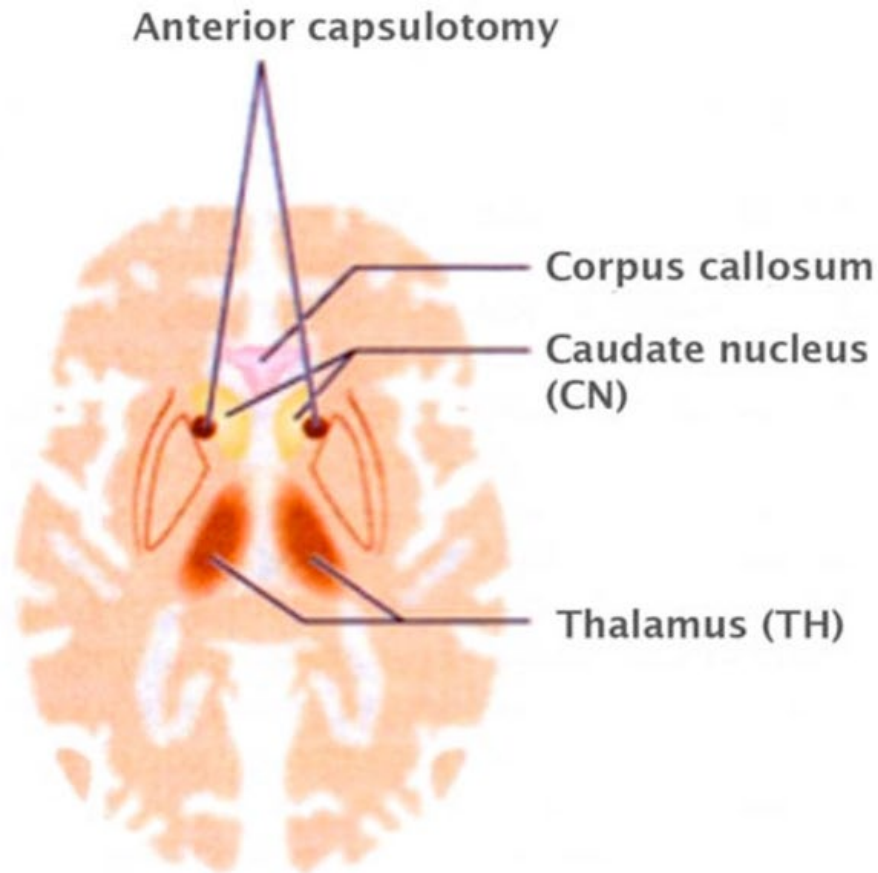
- Cingulotomy
 - Surgical destruction of specific fiber bundles in the subcortical frontal lobe
 - Cingulum bundle-connects the prefrontal and cingulate cortex with the limbic cortex of the temporal lobe



Treatment: OCD

Learning Objective 16.16 Summarize pharmacological treatments for anxiety disorders.

Capsulotomy-caudate nucleus to medial prefrontal cortex bundle



Treatment: OCD

Learning Objective 16.16 Summarize pharmacological treatments for anxiety disorders.

Surgical Treatment

- Side effects
 - Problems of planning, apathy, or difficulty inhibiting socially inappropriate behavior